



Welcome

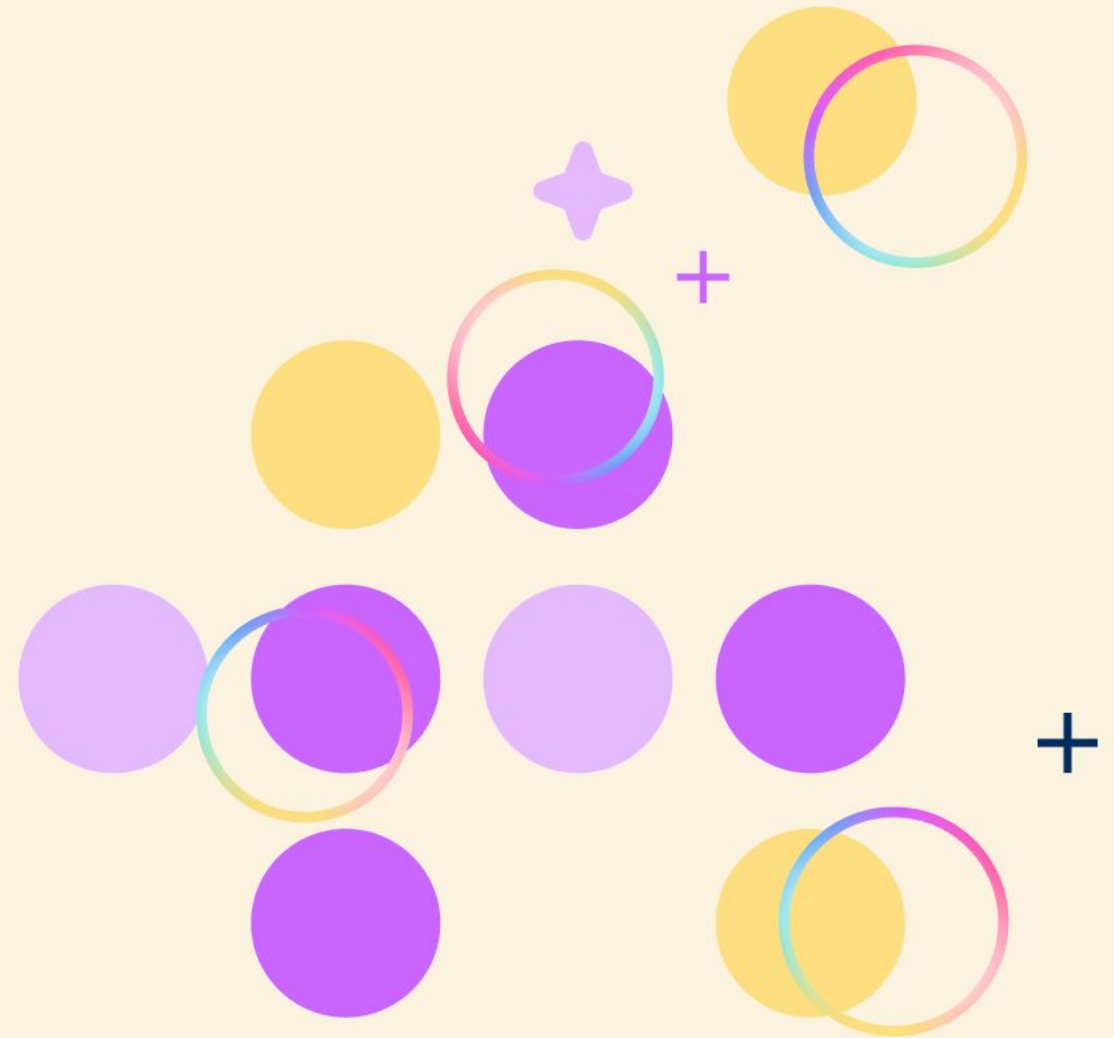
Hands-on Training Room 29B

Wi-Fi: HoT_Room_29B
Password: HotNCold

Get ready:

1. Connect to this room's Wi-Fi.
2. Log into sforce.co/mytc25.
3. Scroll to the Hands-on Training Lab Catalog.
4. Find this session* and select Access Lab.

* After scanning at the door, it may take up to 2 minutes for your lab to appear.





Welcome

Hands-on Training Room 24C

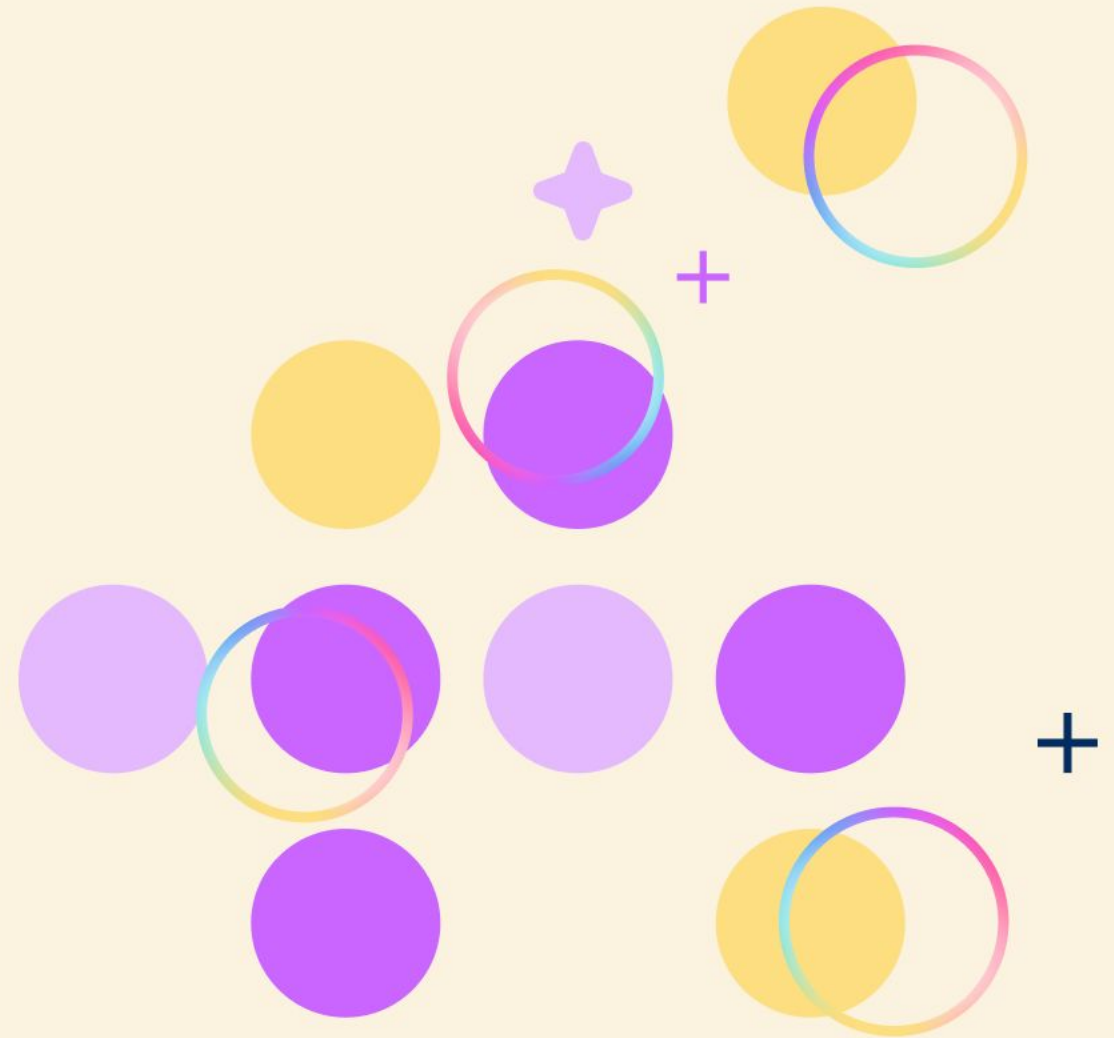
Wi-Fi: HoT_Room_24C

Password: DropItLikeItsHot

Get ready:

1. Connect to this room's Wi-Fi.
2. Log into sforce.co/mytc25.
3. Scroll to the Hands-on Training Lab Catalog.
4. Find this session* and select Access Lab.

* After scanning at the door, it may take up to 2 minutes for your lab to appear.





Welcome

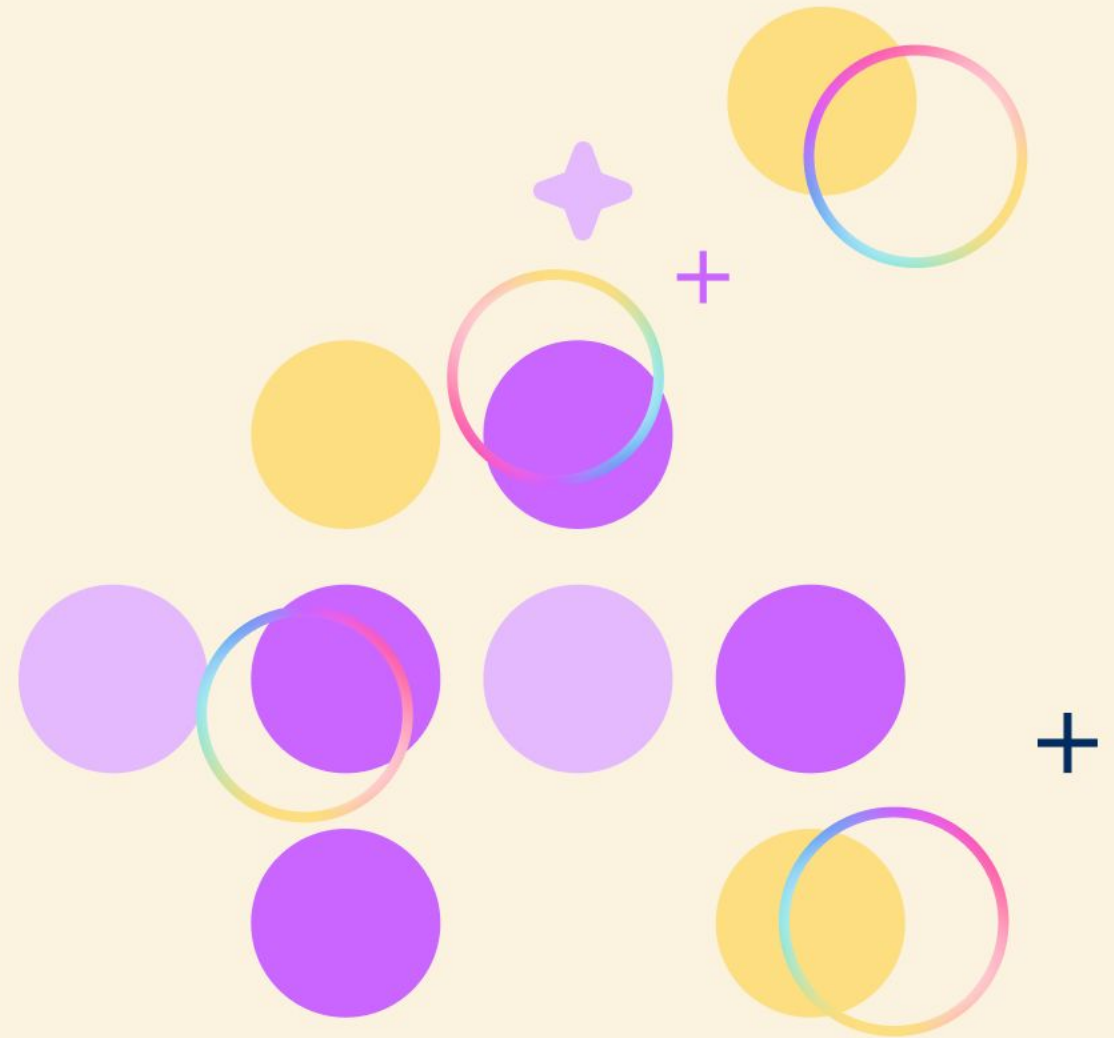
Hands-on Training Room 30C

Wi-Fi: HoT_Room_30C
Password: HotCorner

Get ready:

1. Connect to this room's Wi-Fi.
2. Log into sforce.co/mytc25.
3. Scroll to the Hands-on Training Lab Catalog.
4. Find this session* and select Access Lab.

* After scanning at the door, it may take up to 2 minutes for your lab to appear.





Welcome

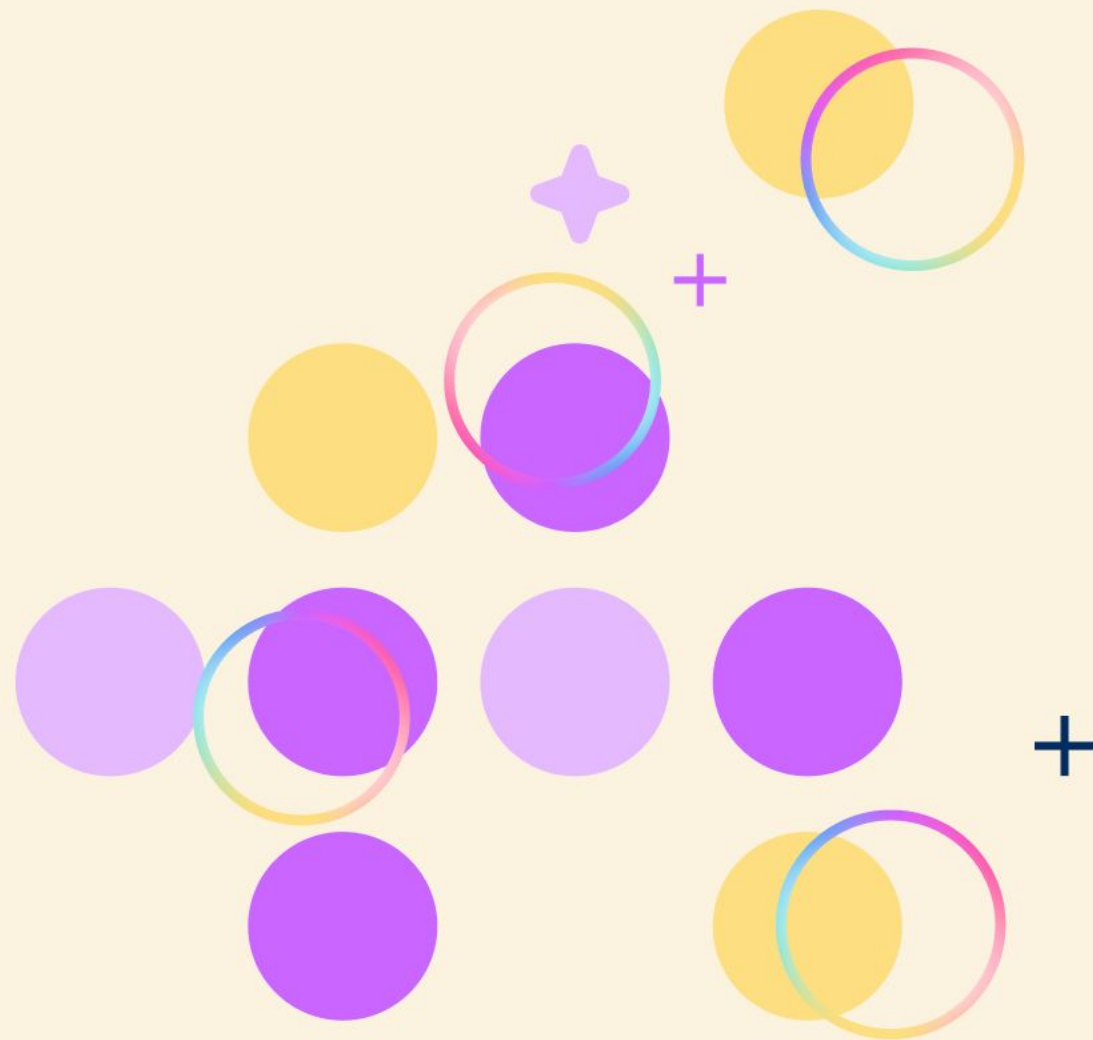
Hands-on Training Room 32A

Wi-Fi: HoT_Room_32A
Password: HotAsIce

Get ready:

1. Connect to this room's Wi-Fi.
2. Log into sforce.co/mytc25.
3. Scroll to the Hands-on Training Lab Catalog.
4. Find this session* and select Access Lab.

* After scanning at the door, it may take up to 2 minutes for your lab to appear.





Tune in using Live Session Audio.

- Scan the QR code or navigate to the Live Session Audio Player.
- Connect headphones to your mobile device.
- Choose this location.
- Begin listening or view captions on your phone.

Salesforce Events Mobile app required.

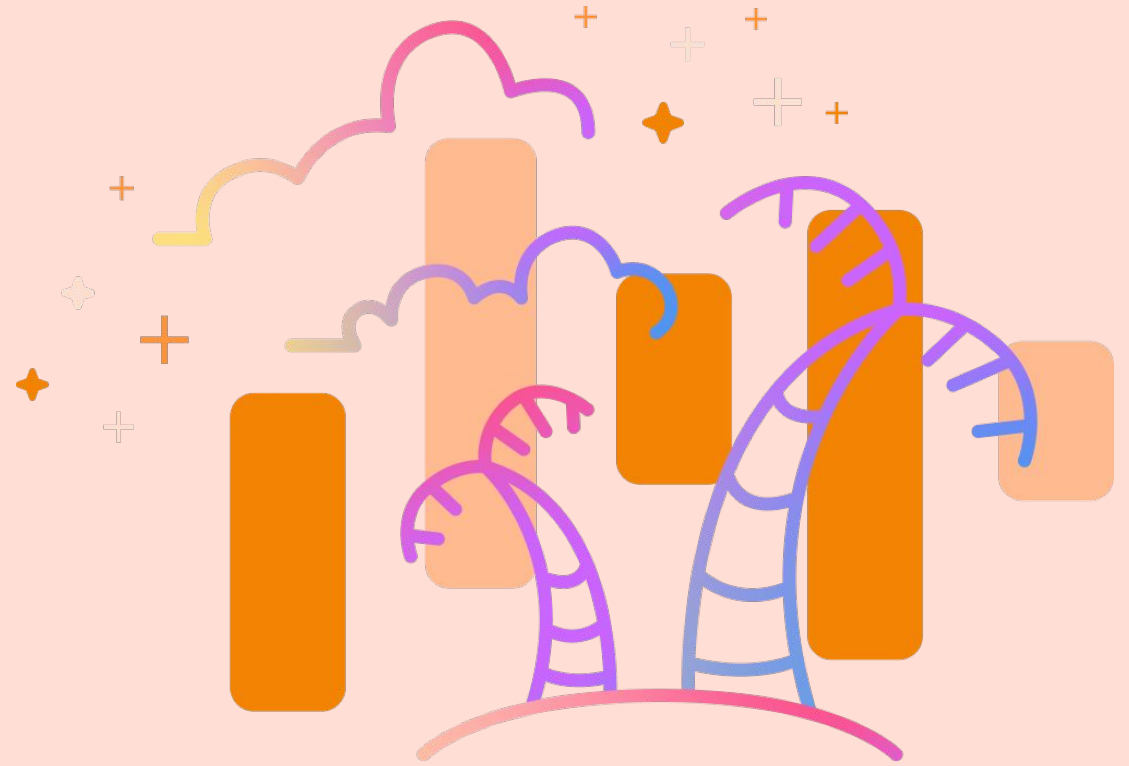


Build, Automate, and Scale:

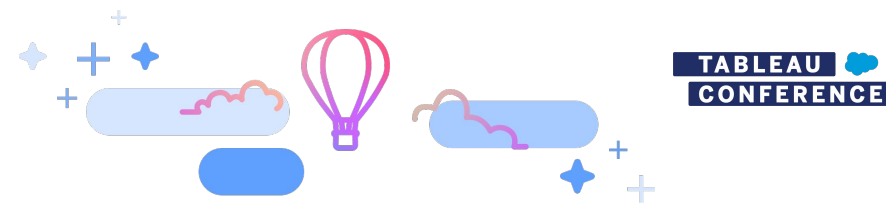
A Tableau REST API Deep Dive

Jeremy Mayo, Principal, Specialist SE

Brian King, Principal, Specialist SE



Forward Looking Statements



This presentation contains forward-looking statements about, among other things, trend analyses and future events, future financial performance, anticipated growth, industry prospects, environmental, social and governance goals, and the anticipated benefits of acquired companies. The achievement or success of the matters covered by such forward-looking statements involves risks, uncertainties and assumptions. If any such risks or uncertainties materialize or if any of the assumptions prove incorrect, Salesforce's results could differ materially from the results expressed or implied by these forward-looking statements. The risks and uncertainties referred to above include those factors discussed in Salesforce's reports filed from time to time with the Securities and Exchange Commission, including, but not limited to: impact of, and actions we may take in response to, the COVID-19 pandemic, related public health measures and resulting economic downturn and market volatility; our ability to maintain security levels and service performance meeting the expectations of our customers, and the resources and costs required to avoid unanticipated downtime and prevent, detect and remediate performance degradation and security breaches; the expenses associated with our data centers and third-party infrastructure providers; our ability to secure additional data center capacity; our reliance on third-party hardware, software and platform providers; the effect of evolving domestic and foreign government regulations, including those related to the provision of services on the Internet, those related to accessing the Internet, and those addressing data privacy, cross-border data transfers and import and export controls; current and potential litigation involving us or our industry, including litigation involving acquired entities such as Tableau Software, Inc. and Slack Technologies, Inc., and the resolution or settlement thereof; regulatory developments and regulatory investigations involving us or affecting our industry; our ability to successfully introduce new services and product features, including any efforts to expand our services; the success of our strategy of acquiring or making investments in complementary businesses, joint ventures, services, technologies and intellectual property rights; our ability to complete, on a timely basis or at all, announced transactions; our ability to realize the benefits from acquisitions, strategic partnerships, joint ventures and investments, including our July 2021 acquisition of Slack Technologies, Inc., and successfully integrate acquired businesses and technologies; our ability to compete in the markets in which we participate; the success of our business strategy and our plan to build our business, including our strategy to be a leading provider of enterprise cloud computing applications and platforms; our ability to execute our business plans; our ability to continue to grow unearned revenue and remaining performance obligation; the pace of change and innovation in enterprise cloud computing services; the seasonal nature of our sales cycles; our ability to limit customer attrition and costs related to those efforts; the success of our international expansion strategy; the demands on our personnel and infrastructure resulting from significant growth in our customer base and operations, including as a result of acquisitions; our ability to preserve our workplace culture, including as a result of our decisions regarding our current and future office environments or work-from-home policies; our dependency on the development and maintenance of the infrastructure of the Internet; our real estate and office facilities strategy and related costs and uncertainties; fluctuations in, and our ability to predict, our operating results and cash flows; the variability in our results arising from the accounting for term license revenue products; the performance and fair value of our investments in complementary businesses through our strategic investment portfolio; the impact of future gains or losses from our strategic investment portfolio, including gains or losses from overall market conditions that may affect the publicly traded companies within our strategic investment portfolio; our ability to protect our intellectual property rights; our ability to develop our brands; the impact of foreign currency exchange rate and interest rate fluctuations on our results; the valuation of our deferred tax assets and the release of related valuation allowances; the potential availability of additional tax assets in the future; the impact of new accounting pronouncements and tax laws; uncertainties affecting our ability to estimate our tax rate; uncertainties regarding our tax obligations in connection with potential jurisdictional transfers of intellectual property, including the tax rate, the timing of the transfer and the value of such transferred intellectual property; uncertainties regarding the effect of general economic and market conditions; the impact of geopolitical events; uncertainties regarding the impact of expensing stock options and other equity awards; the sufficiency of our capital resources; the ability to execute our Share Repurchase Program; our ability to comply with our debt covenants and lease obligations; the impact of climate change, natural disasters and actual or threatened public health emergencies; and our ability to achieve our aspirations, goals and projections related to our environmental, social and governance initiatives.





TABLEAU 
CONFERENCE



Thank you

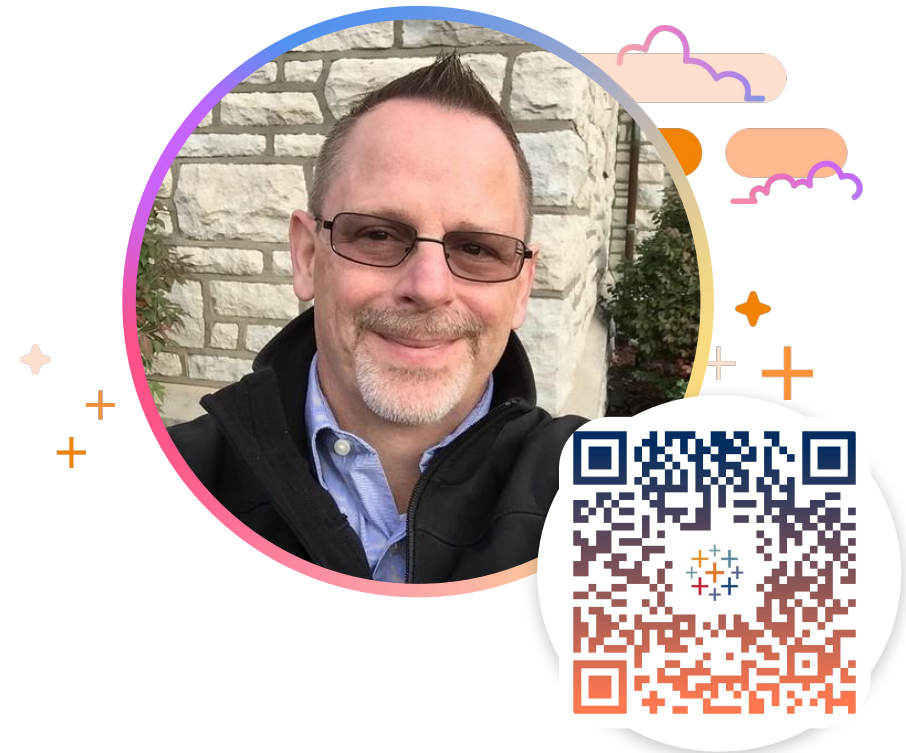


About Us



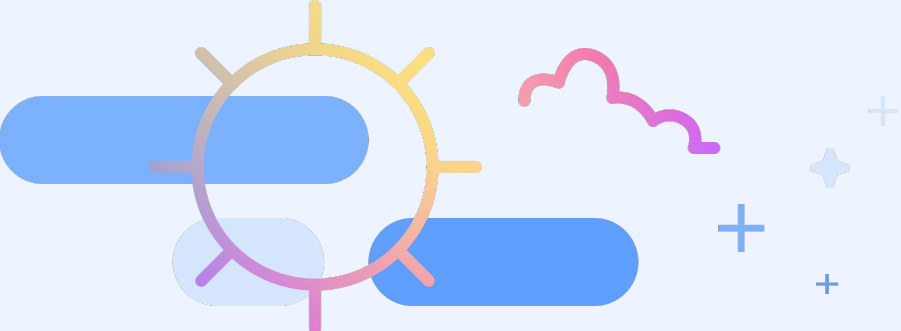
Jeremy Mayo

Principal, Specialist SE

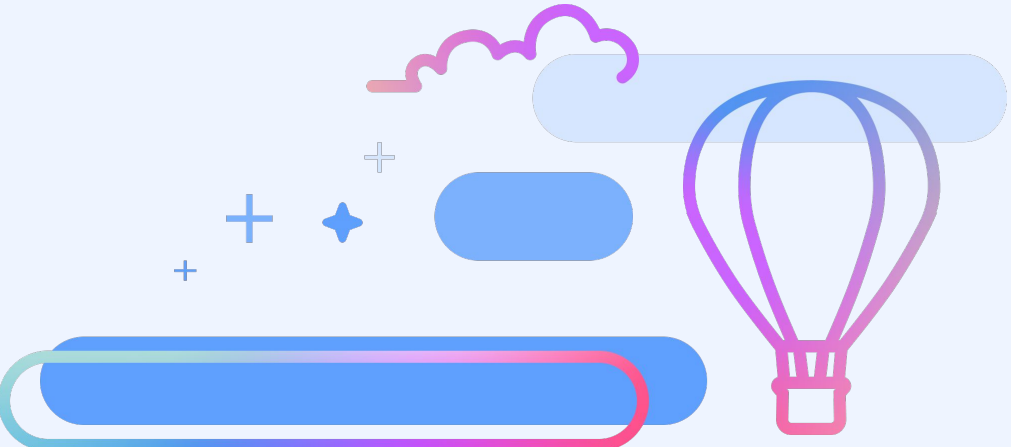


Brian King

Principal, Specialist SE



Agenda



01

Intros, Why are we here?

Quick overview of APIs, Postman, & Tableau

02

Hands-on Training

Using Postman to work with Tableau REST API

03

NEW VizQL Data Service API

Brief description

04

Wrap-up

Q & A



How This Session Will Work

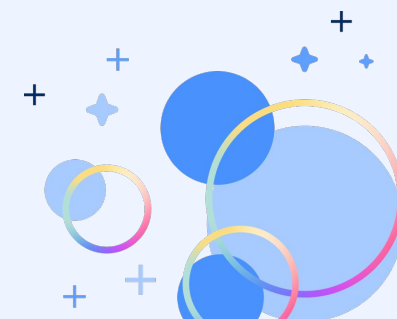
Optional subtitle placeholder

During the Training

- Content in your Lab Studio
 - **Tableau Cloud:** Provided HOT Site
 - **Postman client:** Free account required
- Step by step instructions for all exercises in Lab Manual
- For each topic we'll give some background and then go into hands-on training!

After the Training

- All material will be made available after the session
- Survey at the end, **please give us feedback!**





What is an API?

Why would we use them?



An application programming interface (API) is a way for two or more computer programs to communicate with each other

Wikipedia

<https://en.wikipedia.org/wiki/API>





Nice to see a familiar face



Application



Tableau, Salesforce,
X, Netflix, ChatGPT

Programming



Scripts, Applications,
Agents, Workflows

API

Interface

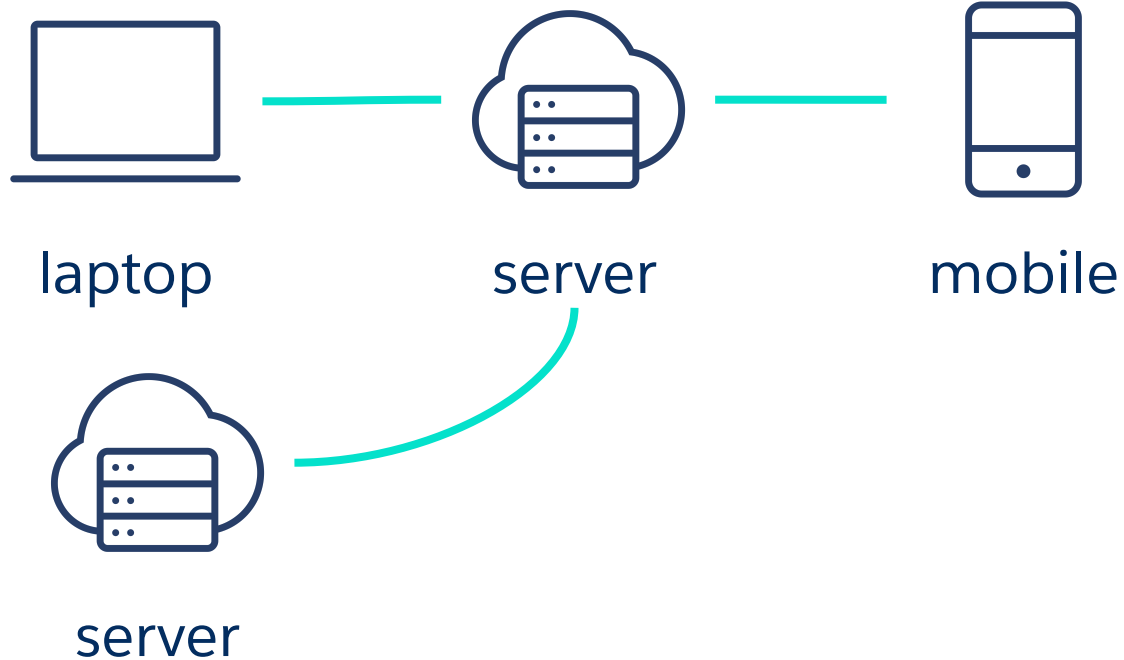


Combine, Collaborate,
Framework, Rules



HTTP is like a conversation

- requests and responses
- data and communication
- rules and protocols



conversation

REpresentational
State
Transfer



HTTP Resources

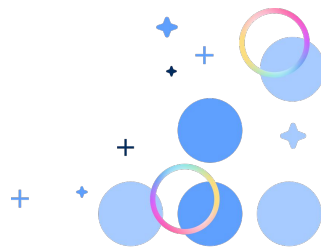
The subjects of the conversation...URLs

/workbooks

/datasources

/users

/projects



HTTP Methods

The verbs of the conversation...CRUD

PUT
UPDATE

DELETE
DELETE

POST
CREATE

GET
READ

Create

Read

Update

Delete



HTTP Response Codes

The emojis of the conversation :)

404
NOT FOUND

401
UNAUTHORIZED

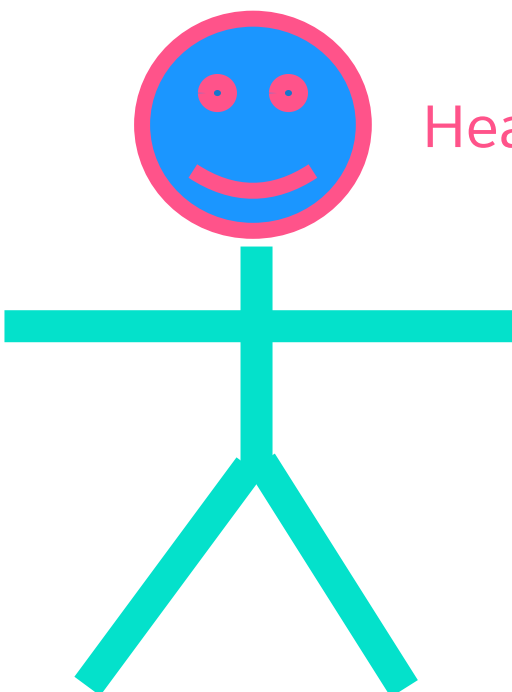
200
SUCCESS

500
INTERNAL ERROR



Anatomy of an HTTP Request

Headers and Bodies



The diagram shows a stick figure with a blue head and red body. The head is labeled "Header" and the body is labeled "Body".

The screenshot displays a web browser's developer console with the "Console" tab selected. The console shows a POST request to `http://win-srv2022/api/3.20/auth/signin`. The "Request Headers" section is expanded, showing the following headers:

- Content-Type: "application/xml"
- User-Agent: "PostmanRuntime/7.37.3"
- Accept: "*/*"
- Cache-Control: "no-cache"
- Postman-Token: "440c0e42-88f4-4aaa-a78a-d267bbec036e"
- Host: "win-srv2022"
- Accept-Encoding: "gzip, deflate, br"
- Connection: "keep-alive"
- Content-Length: "208"

The "Response Headers" section is also expanded, showing the following headers:

- Date: "Mon, 15 Apr 2024 19:15:04 GMT"
- Server: "Tableau"
- Content-Type: "application/xml; charset=utf-8"
- Vary: "Accept-Encoding"
- Content-Encoding: "gzip"
- X-Tableau: "Tableau Server"
- P3P: "CP=NON"
- X-UA-Compatible: "IE=Edge"
- X-Content-Type-Options: "nosniff"
- X-XSS-Protection: "1; mode=block"
- Referrer-Policy: "no-referrer-when-downgrade"
- Content-Security-Policy-Report-Only: "connect-src * https://*.tiles.mapbox.com https://api.mapbox.com; default-src blob;; font-src * data;; frame-src * data: tableau-desktop;; img-src * data: blob;; object-src data;; report-uri /vizql/csp-report; script-src * blob: 'wasm-unsafe-eval'; style-src * 'unsafe-inline'"
- Keep-Alive: "timeout=8, max=100"
- Connection: "Keep-Alive"
- Transfer-Encoding: "chunked"

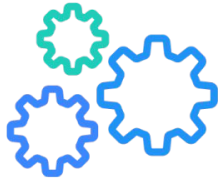
The "Response Body" section is also expanded, showing the following content:

```
"New Session! api-key: msMYB9AAQhy1RTSHClq6Fw|9tKYStRnTfjDeQjsGPTcsiujwVMdgQY|e57ee9d3-bc13-4707-b625-1020dc95c1ab, site-id: e57ee9d3-bc13-4707-b625-1020dc95c1ab"
```

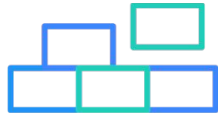


What about Tableau?

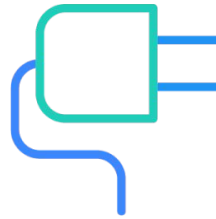
Public Tableau APIs & SDKs



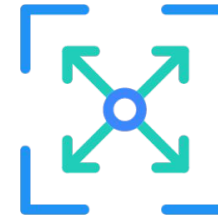
REST
Metadata



Embedding
VizQL Data
Service



Hyper
Connector SDK
Web Data Connection



Dashboard
Extensions



Tabpy
R Integration

Automate

Increase efficiency and productivity
Explore and query metadata about your Tableau content and external data assets

Embed

Quickly integrate the AI-powered, interactive, and analytical features of Tableau into products, applications, and web portals

Connect

Build custom connectors or Tableau extracts with Hyper to bring any data into the Tableau platform

Extend

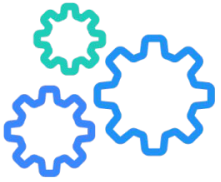
Create new chat types, use agents to automate workflows, integrate third-party functionality, and interact with data applications

Enrich

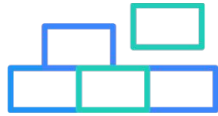
Weave in machine learning code and packages like Python and R and enhance your Tableau AI insights

Public Tableau APIs & SDKs

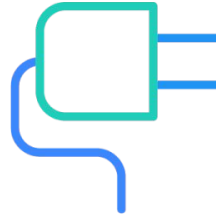
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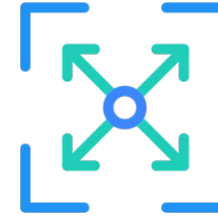
REST
Metadata



Embedding
VizQL Data
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Hyper
Connector SDK
Web Data Connection



Dashboard
Extensions



Tabpy
R Integration

Automate

Increase efficiency and productivity
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Quickly integrate the AI-powered, interactive, and analytical features of Tableau into products, applications, and web portals

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Build custom connectors or Tableau extracts with Hyper to bring any data into the Tableau platform

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Create new chat types, use agents to automate workflows, integrate third-party functionality, and interact with data applications

Enrich

Weave in machine learning code and packages like Python and R and enhance your Tableau AI insights

We have the docs!

Covers all documentation of the APIs, and where to find them.

- [Tableau REST API Help Documentation](#)
- [Tableau Developer Program](#)
- [Tableau Embedded Playbook](#)
- [Tableau Server Client Library](#) - Python wrapper library
- [Tableau API Collection on Postman](#)
- [Get Started Tutorial Part 1: Tools, REST Basics, and Sign In](#)

Methods



Authentication

Access
Impersonation



Users and Groups

CRUD
Governance
Scale



Projects

Organization
Permissions
Multitenancy



Workbooks and Views

CRUD
Expansive API surface

Join #datadev

TABLEAU
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What's a DataDev?

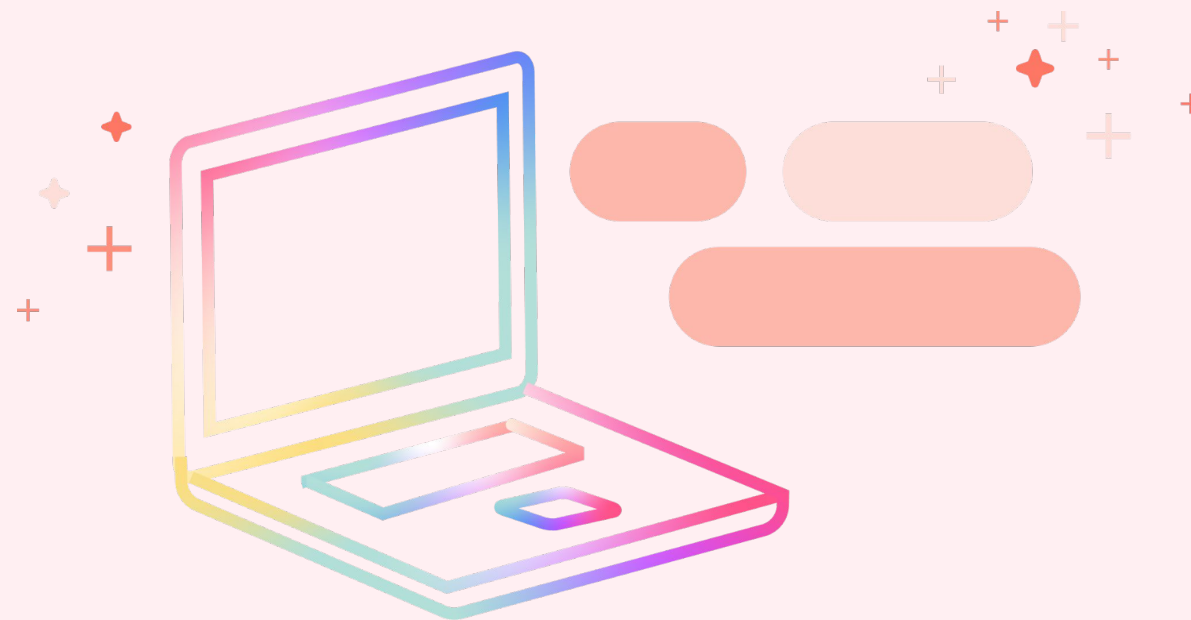
DataDevs are an integral part of the Tableau Community. These innovative minds build, fine-tune, and customize Tableau for their organizations.

The Tableau Developer program

The Tableau Developer Program provides the training, skill building and networking for all levels of DataDevs, from new developers to more seasoned programmers. When you join our developer program, you get your own personal development sandbox, which enables you to test APIs and other developer tools in a safe environment. Be part of the DataDev community and actively shape future Tableau solutions.

[JOIN THE TABLEAU DEVELOPER PROGRAM](#) →



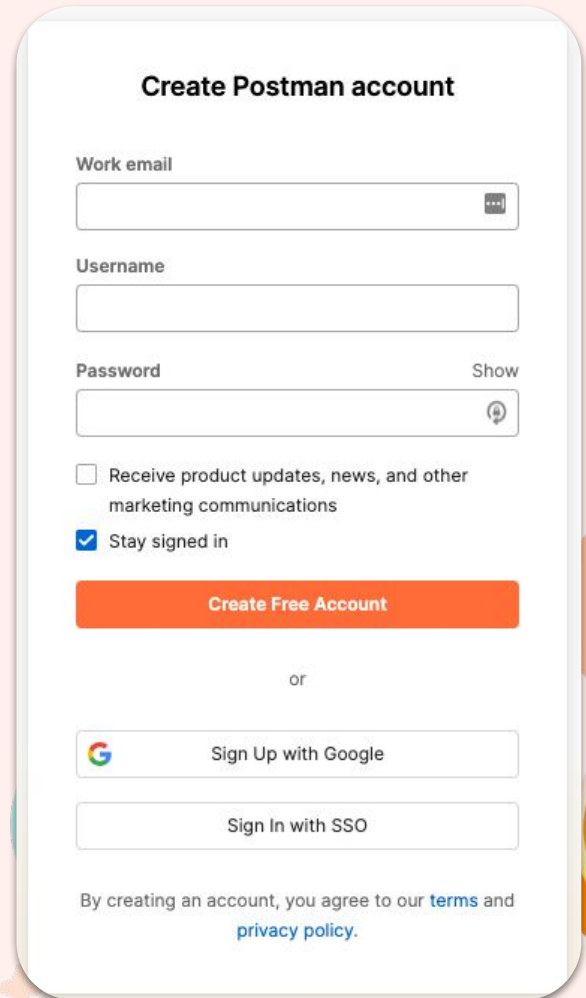


What is Postman?

*** We will be using Postman during this session, which requires an account to use the full application. ***

Please go to
<https://identity.getpostman.com/signup>
and create your free account.

If you already have an account,
fantastic! Please sign into Postman in
your VM.

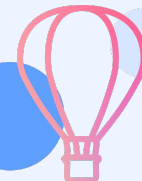
A screenshot of the Postman account creation and login interface. The form is titled "Create Postman account" and includes fields for "Work email", "Username", and "Password". There is a "Show" button next to the password field. Below the fields are two checkboxes: "Receive product updates, news, and other marketing communications" (unchecked) and "Stay signed in" (checked). An orange "Create Free Account" button is positioned below the checkboxes. Below this button is the word "or". There are two more buttons: "Sign Up with Google" (with a Google logo) and "Sign In with SSO". At the bottom, a line of text states: "By creating an account, you agree to our [terms](#) and [privacy policy](#)."



“...a platform for building and using APIs.”

POSTMAN

“...an HTTP client that helps you explore APIs at the speed of thought — much like Tableau helps you analyze data at the speed of thought.”



Use the local app or in the browser

<https://www.postman.com/salesforce-developers/workspace/salesforce-developers/overview>

The screenshot shows the Postman interface with the 'Tableau APIs' collection selected in the left sidebar. The main panel displays the 'Tableau REST API Collection' with a description and a 'Get Started' section.

Tableau APIs

Tableau REST API Collection

The Tableau REST API enables you to perform many Tableau actions from scripts and programs. You can use this collection to make requests to your Tableau site and see the response, without creating an app to do it.

Get Started

To get started, see [How to Use the Collection](#).

Contribute

This collection is open source and we welcome your contributions by [creating a Postman pull](#)

<https://blog.postman.com/the-most-popular-collection-in-the-postman-api-network/>

The most popular collection in the Postman API Network



Joyce

September 1, 2022 · 3 mins

Mirror, mirror on the wall, who has the most popular collection of them all? The [Postman API Network](#) is [the world's largest public API hub](#). But in this popularity contest, there's only one measure to determine the winning collection—the one with the most [forks](#).

As of the writing of this blog post, the most popular collection in the Postman API Network belongs to Salesforce. Check out their [Salesforce Platform APIs](#) collection with more than 70,000 forks.

sfcore Platform APIs 

 Run  Fork | 235k  Watch | 116k  Share

view Authorization • Scripts • Variables •

Salesforce Platform APIs

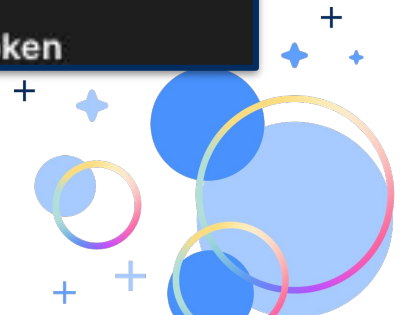
Salesforce Platform APIs collection contains 250+ requests and sample responses for the following APIs:

Creator
 Salesforce Developers 

Category
 Developer Productivity

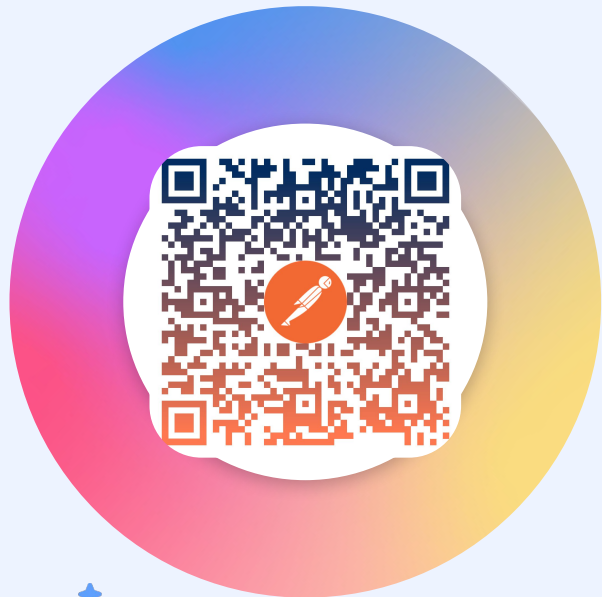
Popular requests
POST JWT Bearer Token

WOW!



Postman

The 2025 Developers' Choice Awards



The 2025 Developers' Choice Awards Live Leaderboard

Which APIs GET you? Vote for the APIs that really make a difference and the top 5 will win the recognition they've earned — from you, the developers who know best. [Learn more](#)

Voting ends Thursday, April 17th



In the lead

1st place

307 votes



Cisco Meraki is a cloud managed IT company headquartered in San Francisco, California. Their solutions include wireless, switching,...

[Vote for Cisco Meraki](#)

2nd place

285 votes



BoldSign APIs are blazing fast and let you embed the entire eSignature process into your application seamlessly.

[Vote for BoldSign](#)

3rd place

251 votes



API resources for Salesforce Developers

✓ You voted for Salesforce Developers

4th place

199 votes



Alpaca's APIs allow anyone to build and connect applications or algorithms to trade stocks, options and crypto.

[Vote for Alpaca](#)

5th place

177 votes



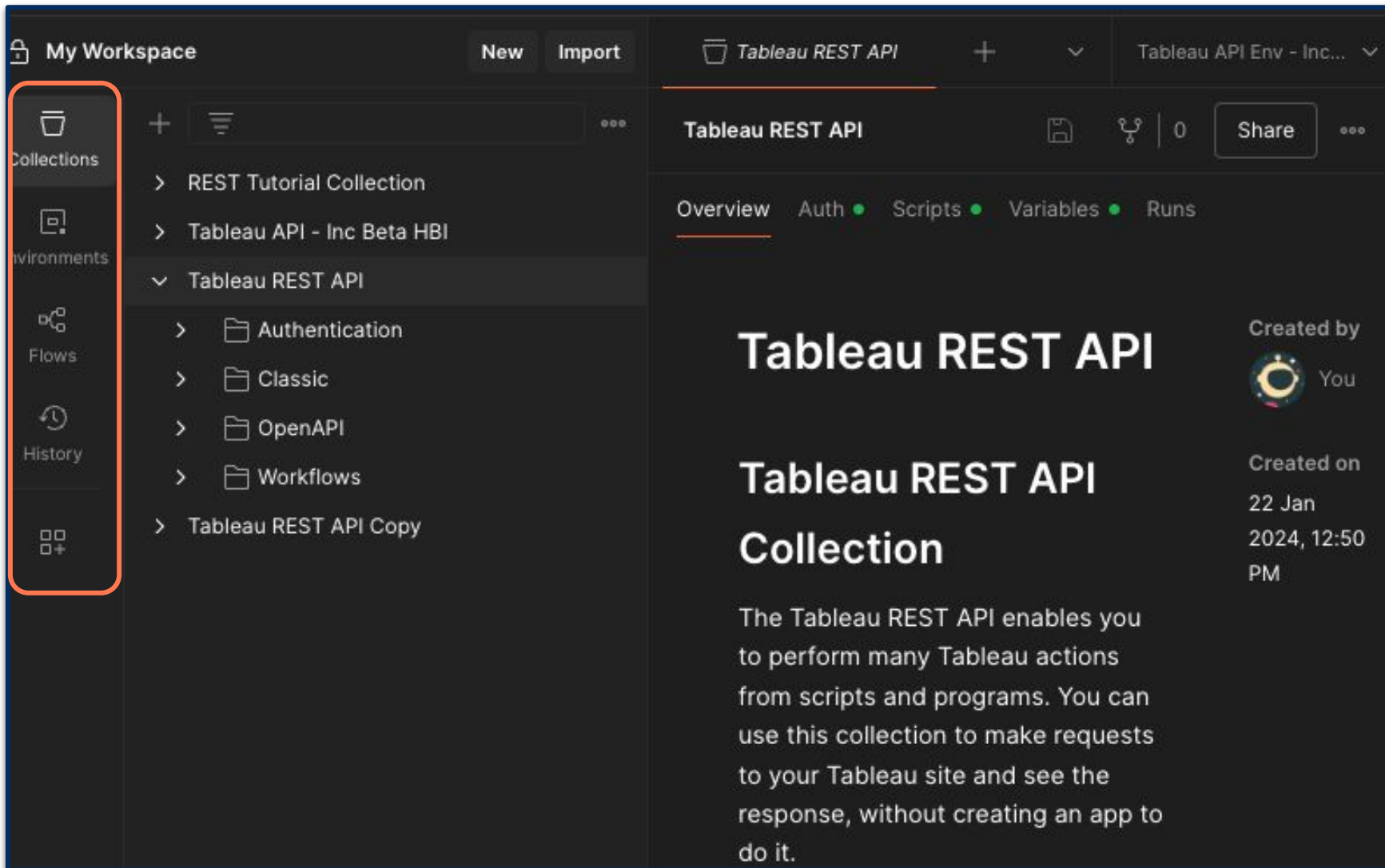
PandaDoc is the easy and scalable all-in-one API solution for Document Generation and eSignatures. Get started with fully featured...

[Vote for PandaDoc](#)



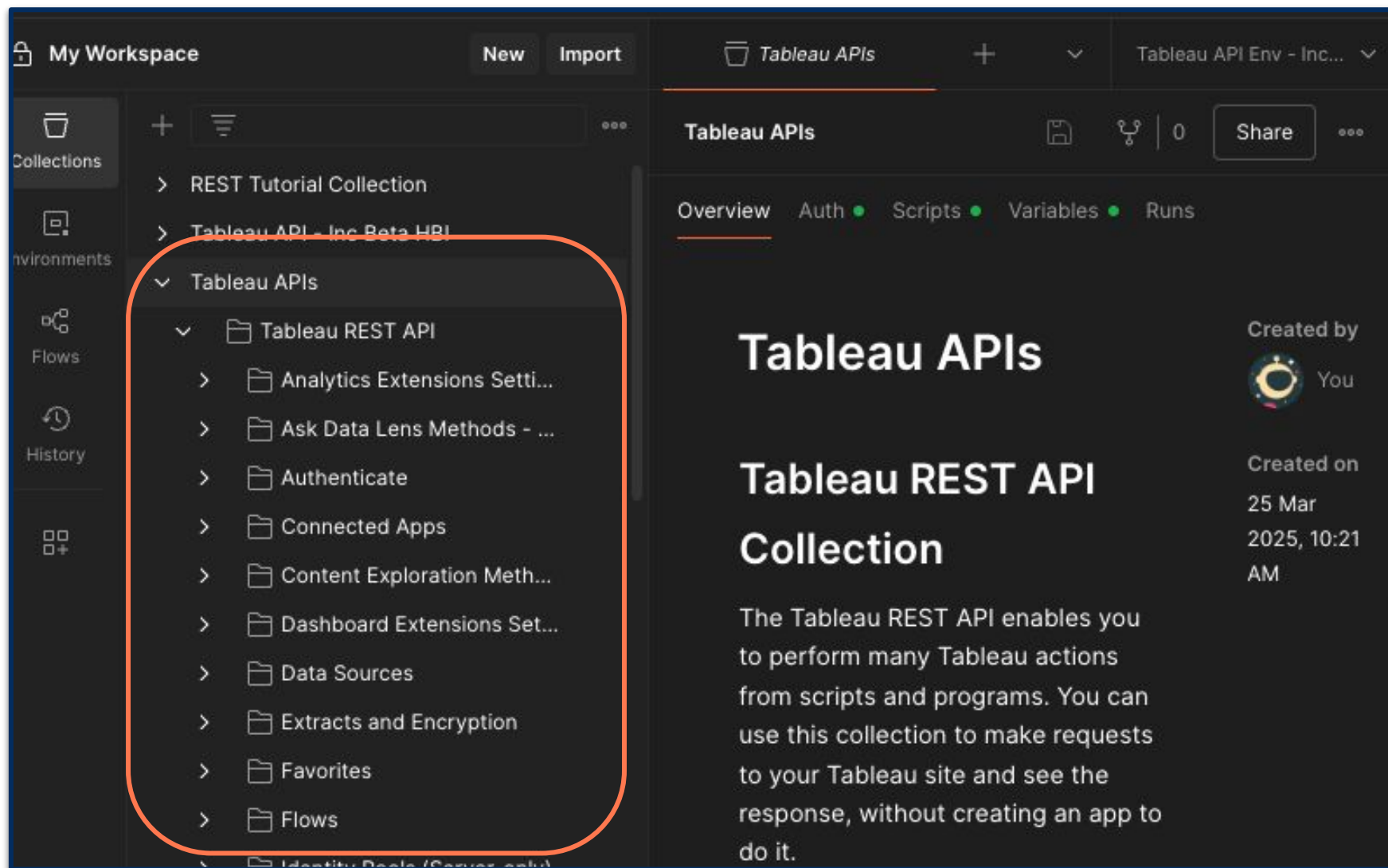
The Left Sidebar

Fundamental elements or sections in the app



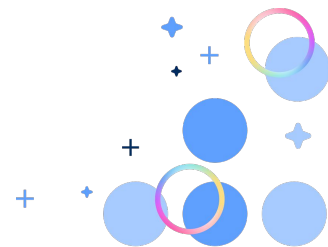
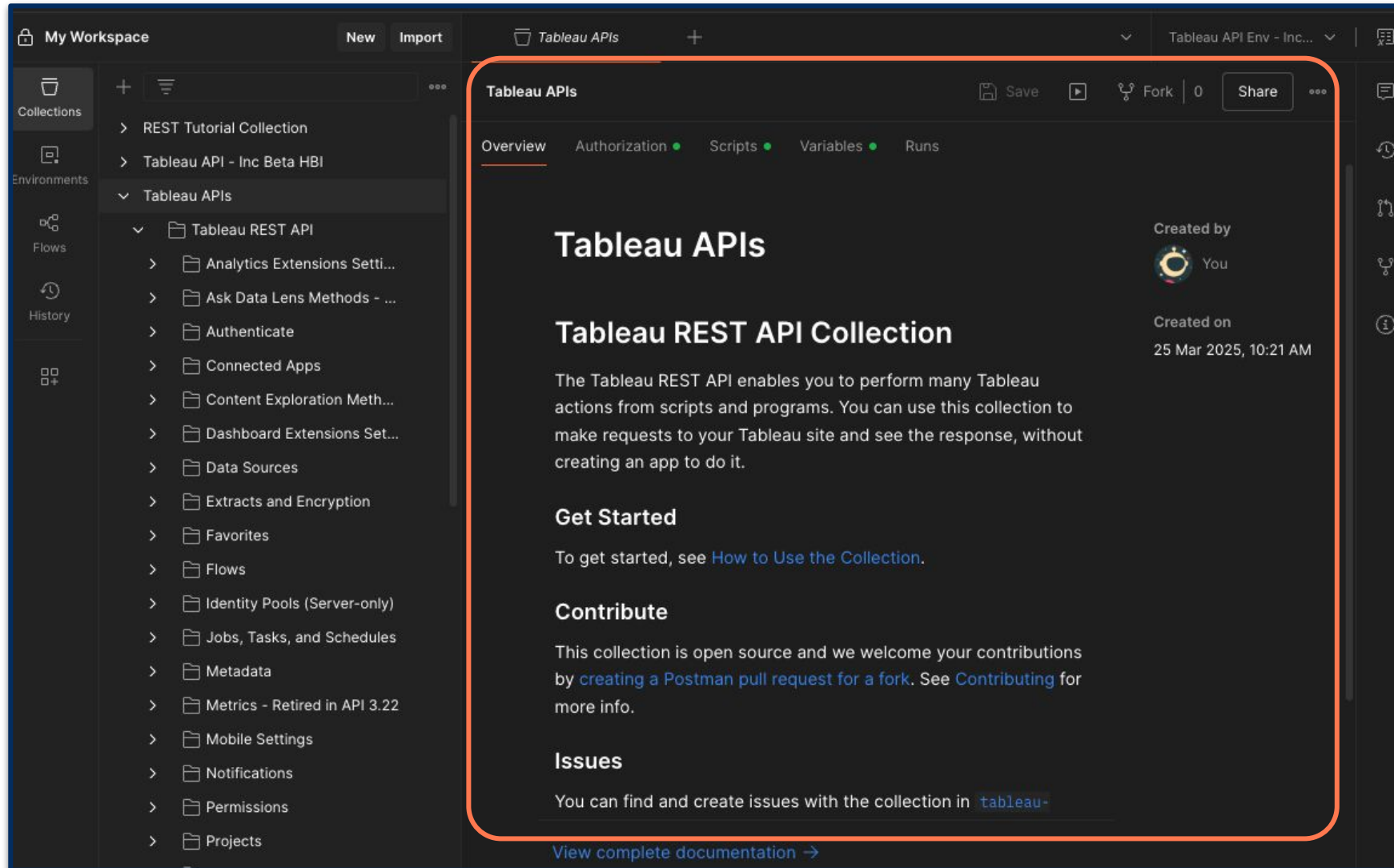
Directory

Organizes collection methods using folders



Workbench

The main area of action



Anatomy of the Workbench

Headers and bodies for requests and responses

The screenshot displays the Tableau REST API client interface. On the left, a sidebar lists various API endpoints under 'Tableau APIs' and 'Authenticate'. The main panel shows a POST request to `{{base-path}}/auth/signin`. The request body is in XML format, enclosed in a cyan box, and contains a `<tsRequest>` element with `<credentials>` and `<site contentUrl>` sub-elements. The word 'REQUEST' is written in cyan next to it. Below the request, the response is shown in a yellow box. The status bar indicates a '200 OK' status code, highlighted with a blue box and labeled 'STATUS CODE' in blue. The response body is in XML format and contains a `<tsResponse>` element with a `<credentials>` sub-element containing a token. The word 'RESPONSE' is written in yellow next to it.

REQUEST

```
1 <tsRequest>
2   <credentials jwt="{{jwt}}">
3     <site contentUrl="{{content-url}}" />
4   </credentials>
5 </tsRequest>
```

STATUS CODE → 200 OK

RESPONSE

```
1 <?xml version='1.0' encoding='UTF-8'?>
2 <tsResponse xmlns="http://tableau.com/api" xmlns:xsi="http://www.w3.org/2001/
  XMLSchema-instance" xsi:schemaLocation="http://tableau.com/api https://help.
  tableau.com/samples/en-us/rest_api/ts-api_3_20.xsd">
3   <credentials token="oeXBwF8TRemXRD04aQNu7w|6pij40ob7g9S7HvzNQtWSwoxvGrNZrLb|
    b9b71e5c-be62-40e8-8791-2dc97fe71013">
```

The Right Sidebar

Additional tools such as code generation, docs and contributions

The screenshot displays the Tableau API Network interface. The top navigation bar includes 'Home', 'Workspaces', and 'API Network'. The main content area shows a collection of 'Tableau APIs' with tabs for 'Overview', 'Authorization', 'Scripts', 'Variables', and 'Runs'. The 'Overview' tab is active, displaying the title 'Tableau APIs' and 'Tableau REST API Collection'. Below the title, a description states: 'The Tableau REST API enables you to perform many Tableau actions from scripts and programs. You can use this collection to make requests to your Tableau site and see the response, without creating an app to do it.' The right sidebar, highlighted with an orange border, contains several icons: a document with a checkmark, a speech bubble, a circular arrow, a fork icon, a share icon, and an information icon. The sidebar also shows the user's profile and the creation date and time of the collection.

TC25 REST API HOT

New Import

Tableau APIs

Tableau REST API En...

Upgrade

Collections

Environments

Flows

History

Tableau APIs

Tableau REST API

Analytics Extensions Setti...

Ask Data Lens Methods - ...

Authenticate

Connected Apps

Content Exploration Meth...

Dashboard Extensions Set...

Data Sources

Extracts and Encryption

Favorites

Flows

Tableau APIs

Save Fork 0 Watch 0 Share

Overview Authorization Scripts Variables Runs

Tableau APIs

Tableau REST API Collection

The Tableau REST API enables you to perform many Tableau actions from scripts and programs. You can use this collection to make requests to your Tableau site and see the response, without creating an app to do it.

Created by You

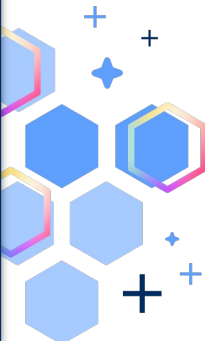
Created on 28 Mar 2025, 11:23 AM

Environment Files

Stores variables used to describe a Tableau site or environment

The screenshot displays the Tableau REST API Environment interface. On the left sidebar, the 'Environments' tab is selected. The main panel shows a list of environments, with 'Tableau REST API Environment' highlighted. The right panel displays the configuration for this environment, including a table of variables.

	Variable	Type	Initial value	Current value
☒	server	defa...	prod-uswest-c.onli...	prod-uswest-c.online.tablea...
☒	api-version	defa...	3.25	3.25
☒	content-url	defa...	tcrestapihot	tcrestapihot
☒	site-id	defa...	b9b71e5c-be62-40...	b9b71e5c-be62-40e8-8791...
☒	site-name	defa...		tcrestapihot
☒	admin-username	secret		
☒	admin-password	secret		
☒	admin-PAT-name	secret		
☒	admin-PAT-secret	secret		
☒	user-username	secret		
☒	user-password	secret		
☒	user-PAT-name	secret		
☒	user-PAT-secret	secret		
☐	jwt	defa...		



Working Environment

Confirm that your environment is active

1. Set Environment

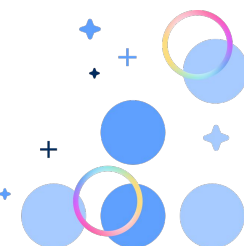
2. Inspect

The screenshot shows the Tableau REST API Environment configuration interface. The left sidebar contains 'My Workspace' with 'New' and 'Import' buttons, and a list of environments: 'New Environment', 'Tableau API Env - Inc Beta HBI', 'Tableau REST API Environment' (selected with a checkmark), and 'Tableau Server (main)'. The main area displays a table of variables for the selected environment. The 'Tableau REST API Environment' dropdown is highlighted with an orange box and an arrow labeled '1. Set Environment'. The 'Inspect' icon (a magnifying glass over a document) is also highlighted with an orange box and an arrow labeled '2. Inspect'. The 'All variables' panel is open, showing a list of variables with their current values.

Variable	T...	Initial ...	Current ...
☒ server	d...	prod-u...	prod-...
☒ api-ver...	d...	3.25	3.25
☒ conten...	d...	tcresta...	tcrest...
☒ site-id	d...	b9b71e...	b9b71...
☒ site-na...	d...		tcrest...
☒ admin-...	s...		
☒ admin-...	s...		
☒ admin-...	s...		
☒ admin-...	s...		
☒ admin-...	s...		
☒ user-u...	s...		•...
☒ user-p...	s...		
☒ user-P...	s...		
☒ user-P...	s...		

All variables

- Environment
- server: prod-uswest-c.onlin...
- api-version: 3.25
- content-url: tcrestapihot
- site-id: b9b71e5c-be62-40e...
- site-name: tcrestapihot
- admin-username: Enter value
- admin-password: Enter value
- admin-PAT-name: Enter value
- admin-PAT-secret: Enter value
- user-username:
- user-password: Enter value
- user-PAT-name: Enter value
- user-PAT-secret: Enter value



Working Environment

Confirm that your environment is active

1. Set Environment

2. Inspect

3. Environment Variables

The screenshot displays the Tableau REST API Environment interface. On the left, the 'My Workspace' sidebar shows a list of environments: 'New Environment', 'Tableau API Env - Inc Beta HBI', 'Tableau REST API Environment' (selected with a checkmark), and 'Tableau Server (main)'. The main panel shows the 'Tableau REST API Environment' configuration. A table lists variables with checkboxes, variable names, types, and initial/current values. An 'All variables' panel is open on the right, showing a list of variables and their values. A red box highlights the 'Tableau REST API Environment' dropdown in the top right, and a red arrow points to the 'Inspect' icon (a magnifying glass over a document) next to it. A red arrow also points to the 'Environment' variable in the 'All variables' panel.

Variable	T...	Initial ...	Current ...
☒ server	d...	prod-u...	prod-...
☒ api-ver...	d...	3.25	3.25
☒ conten...	d...	tcresta...	tcrest...
☒ site-id	d...	b9b71e...	b9b71...
☒ site-na...	d...		tcrest...
☒ admin-...	s...		
☒ admin-...	s...		
☒ admin-...	s...		
☒ admin-...	s...		
☒ admin-...	s...		
☒ user-u...	s...		•...
☒ user-p...	s...		
☒ user-P...	s...		
☒ user-P...	s...		

Variable	Value
Environment	
server	prod-uswest-c.onlin...
api-version	3.25
content-url	tcrestapihot
site-id	b9b71e5c-be62-40e...
site-name	tcrestapihot
admin-username	Enter value
admin-password	Enter value
admin-PAT-name	Enter value
admin-PAT-secret	Enter value
user-username	
user-password	Enter value
user-PAT-name	Enter value
user-PAT-secret	Enter value



Authentication

Authentication

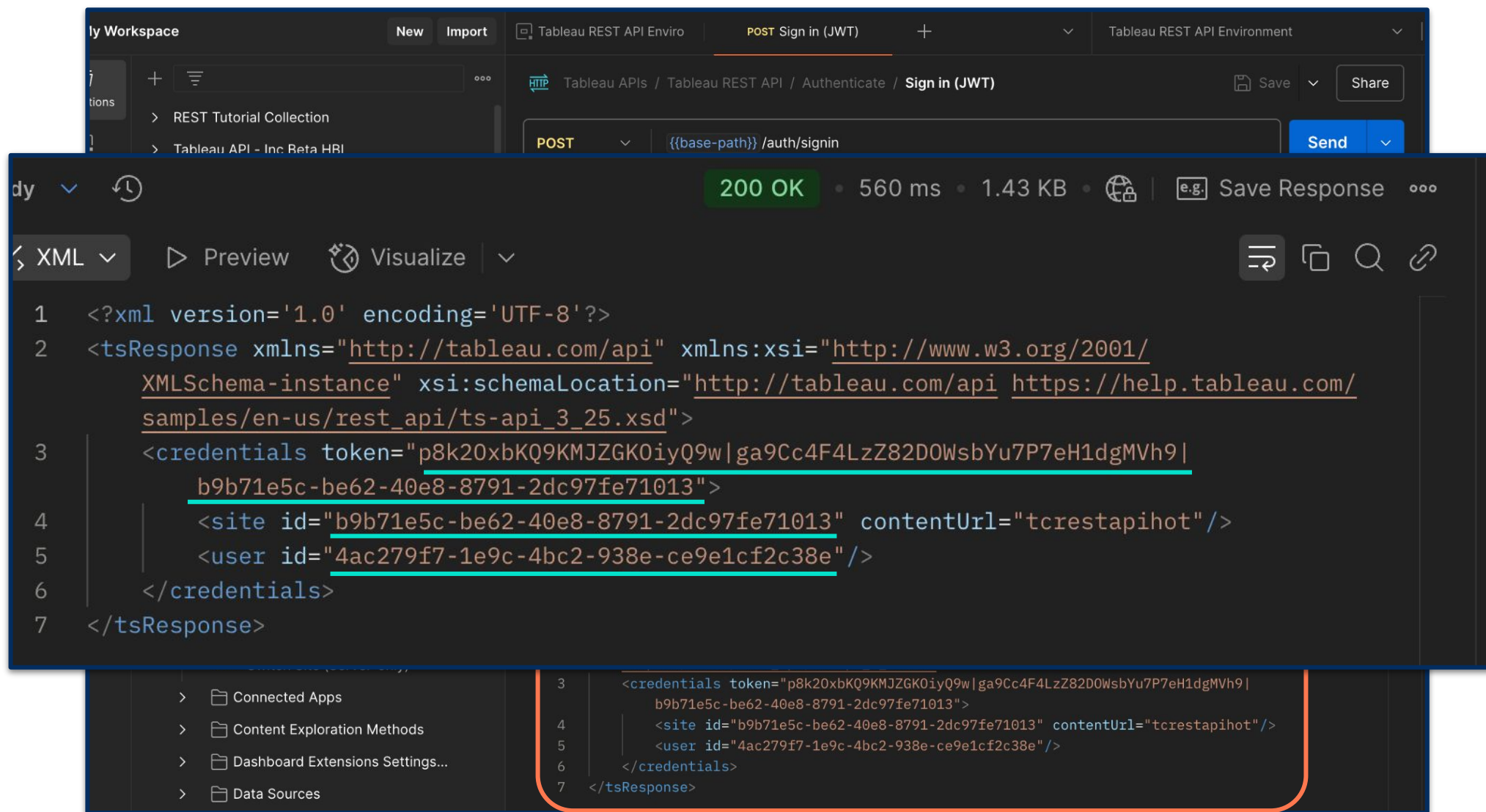
Sign in with a JSON Web Token (JWT)

The screenshot displays the Tableau REST API client interface. On the left sidebar, the 'Collections' section is expanded, showing a tree structure: 'Tableau APIs' > 'Tableau REST API' > 'Authenticate'. The 'Authenticate' folder is selected, and the 'POST Sign in (JWT)' endpoint is highlighted. The main panel shows the configuration for this endpoint: the method is 'POST', the URL is '{{base-path}}/auth/signin', and the 'Body' tab is active. The body content is a JSON Web Token (JWT) request, displayed in raw XML format. The 'Send' button is visible in the top right of the main panel.

```
1 <tsRequest>
2   <credentials jwt="{{jwt}}">
3     <site contentUrl="{{content-url}}" />
4   </credentials>
5 </tsRequest>
```

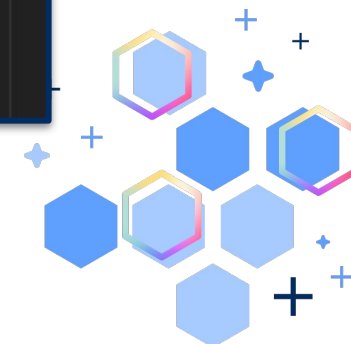
We're In!

The token is the API key or a user session



The screenshot displays the Tableau REST API client interface. The top bar shows the workspace name 'ly Workspace' and the selected environment 'Tableau REST API Environment'. The main area shows a POST request to the endpoint `{{base-path}}/auth/signin`. The response status is **200 OK** with a response time of 560 ms and a size of 1.43 KB. The response is displayed in XML format, showing a `<tsResponse>` element containing a `<credentials>` element with a `token` attribute. The token value is `p8k20xbKQ9KMJZGK0iyQ9w|ga9Cc4F4LzZ82D0WsbYu7P7eH1dgMVh9|b9b71e5c-be62-40e8-8791-2dc97fe71013`. The response also includes `<site>` and `<user>` elements with their respective IDs.

```
<?xml version='1.0' encoding='UTF-8'?>
<tsResponse xmlns="http://tableau.com/api" xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance" xsi:schemaLocation="http://tableau.com/api https://help.tableau.com/
samples/en-us/rest_api/ts-api_3_25.xsd">
  <credentials token="p8k20xbKQ9KMJZGK0iyQ9w|ga9Cc4F4LzZ82D0WsbYu7P7eH1dgMVh9|
  b9b71e5c-be62-40e8-8791-2dc97fe71013">
    <site id="b9b71e5c-be62-40e8-8791-2dc97fe71013" contentUrl="tcrestapihot"/>
    <user id="4ac279f7-1e9c-4bc2-938e-ce9e1cf2c38e"/>
  </credentials>
</tsResponse>
```



Working Environment

Confirm that your environment is active

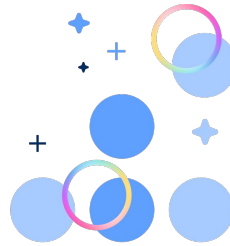
The screenshot displays the Tableau REST API Environment interface. On the left, the 'My Workspace' sidebar shows a tree view of collections and environments. The main panel shows a 'POST Sign in (JWT)' request with a 200 status code. The response body is XML. On the right, the 'Variables in request' panel is open, showing a list of variables. A red box highlights the 'Variables in request' panel, and a yellow box highlights the 'Collection' section within it. An orange arrow points to the '1. Inspect' text, and a yellow arrow points to the '2. Collection Variables' text.

Variables in request

representation_options	Enter value
insights_options	Enter value
jwt_secret	
jwt_secret_id	
jwt_client_id	
c Collection	
auto-auth	false
impersonation	false
protocol	https
base-path	{{protocol}}://{{server...}}
vizql-base-path	{{new-path}}/api/{{viz...}}
new-path	{{protocol}}://{{server}}
api-key	p8k2OxbKQ9KMJZG...
jwt	eyJ0eXAiOiJKV1QiLC...
baseUrl	{{protocol}}://{{server}}

1. Inspect

2. Collection Variables



Quality of Life! X-Tableau-Auth

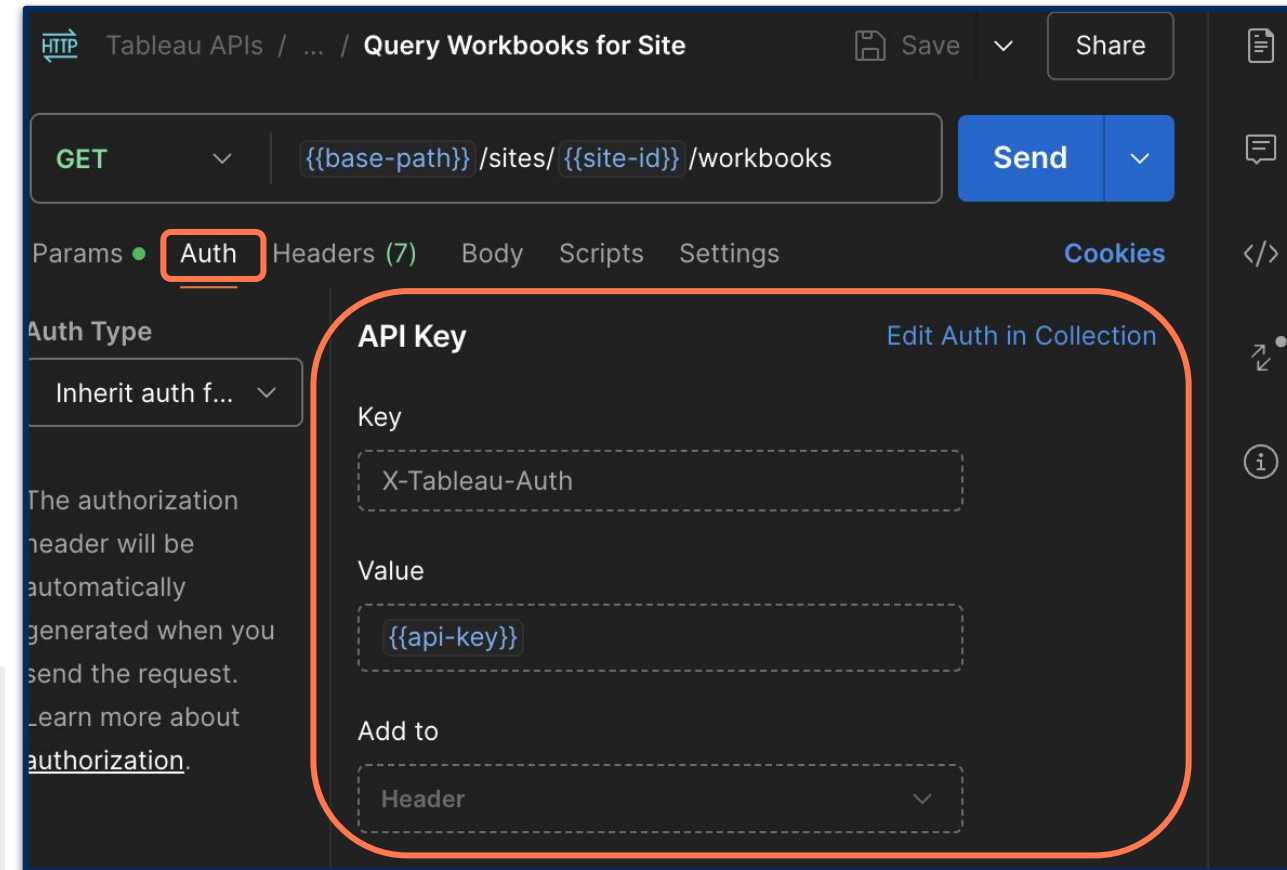
The Collection stores the API key as a variable `{{api-key}}` and uses it on subsequent calls...automatically!

Sign In

Signs you in as a user on the specified site on Tableau Server or Tableau Cloud.

This call returns a credentials token that you use in subsequent calls to the server. Typically, a credentials token is valid for 240 minutes. With administrator permissions on Tableau Server, you can change this timeout by using the `tsm configuration set` command and setting the `wgserver.session.idle_limit` option.

Note: The token is valid for REST API calls and Tableau Metadata API (GraphQL) queries. You cannot use the token as authentication for other operations with Tableau Server. In addition, the token is good only for operations in the site that you're signed in to. You cannot sign in to one site and then use the token you get back to send requests to a different site. If you do, the server returns an HTTP **403 (Forbidden)** error.



*Sources:

https://help.tableau.com/current/api/rest_api/en-us/REST/rest_api_concepts_auth.htm#using_auth_token
https://help.tableau.com/current/api/rest_api/en-us/REST/rest_api_ref_authentication.htm#sign_in



Code Generation - get the X-Tableau-Auth token

Find code gen on the right sidebar

The screenshot shows the Tableau Developer Tools interface. On the left sidebar, the 'Query Workbooks for Site' endpoint is selected and highlighted with a red box and the number 1. In the main panel, the 'Send' button is highlighted with a red box and the number 2. Below it, the 'Cookies' tab is selected with a red box and the number 3. On the right sidebar, the 'Code snippet' panel is open, showing a cURL command. A red box with the number 4 highlights the 'Copy snippet' button. A large red arrow points from the 'Copy snippet' button to the terminal at the bottom of the slide.

1 GET Query Workbooks for Site

2 Send

3 Cookies

4 Copy snippet

Code snippet

```
cURL
1 curl --location 'https://prod-uswest-c.online.tableau.com/api/3.25/sites/b9b71e5c-be62-40e8-8791-2dc97fe71013/workbooks' \
2 --header 'X-Tableau-Auth: .....'
```

Reusable
Session
Token!

```
1 curl --location 'https://prod-uswest-c.online.tableau.com/api/3.25/sites/b9b71e5c-be62-40e8-8791-2dc97fe71013/workbooks'
2 --header 'X-Tableau-Auth: p8k20xbKQ9KMJXGK0iyQ9w|ga9Cc4FLzZ82D0wsbYu7p7eH1dgMVh9|b9b71e5c-be62-40e8-8791-2dc97fe71013'
```

Methods for Authentication

Username and Password

- Fixed to a specific user ⚠
- Credentials sent in the clear payload

Personal Access Token (PAT)

- More secure than U/P
- Still fixed to a specific user ⚠
- PATs expire after 15 days of no use

JSON Web Token via Connected Apps (JWT)

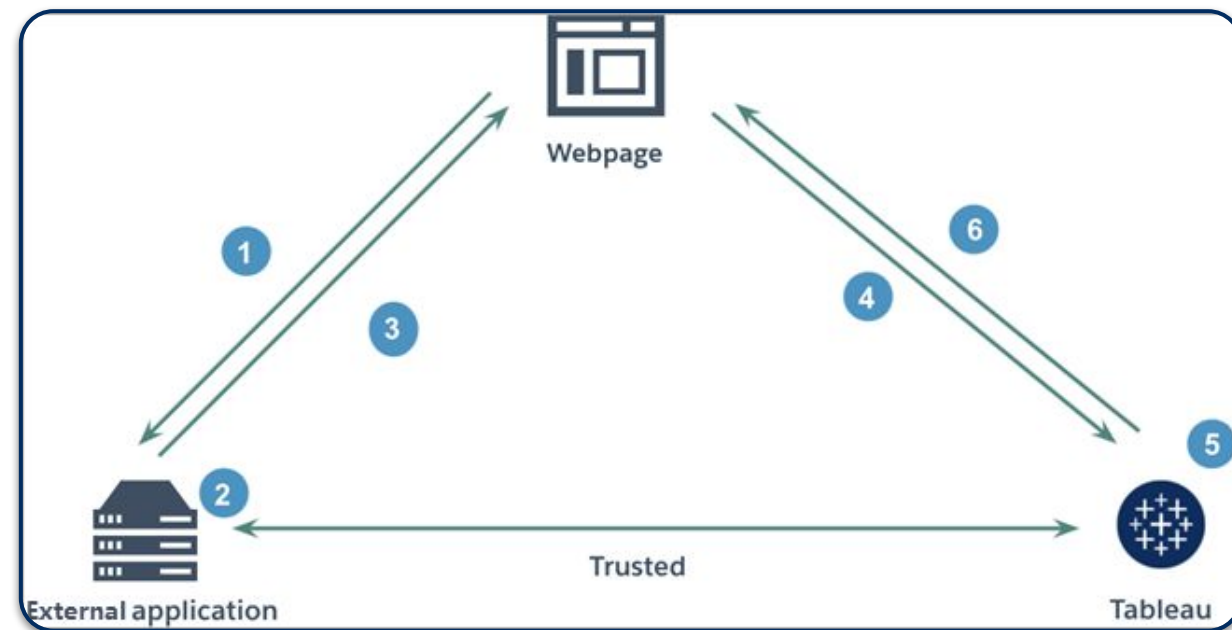
- Modern approach to Authentication
- Session can be scoped to a user's role capabilities
- We're using this during this session!
- Not all methods supported... yet. ⚠



Wait... Connected Apps?

- Launched in January of 2022 for the **Embedding API v3** rewrite
- Also a means to programmatically access the **Tableau REST API** on behalf of a user
- Uses a **shared secret** between your application and Tableau
- Requires **Scopes** (think permissions) that authorizes the user to perform certain actions

["tableau:content:read",
"tableau:datasources:create",
"tableau:datasources:update",
"tableau:datasources:download",
"tableau:tasks:run"]



How to Create a Connected App

1. Goto Settings -> **Connected Apps**
2. Click New Connected App -> **Direct Trust**
3. Give it a **name**
4. Specify a **project** or **all projects**
5. Specify **all domains** or a **specific domain**
6. Click **Create**
7. Then **Generate New Secret**
8. Finally, remember to **Enable**

TC25_test ...

Status **Enabled** Created **Mar 10, 2025**

Secret (Maximum of 2)	Generated on March 10, 2025	Actions ▼
	ID 580c31cc-d256-471c-83d0-857baa29f8f8	
	Value *****	👁

Generate New Secret

Client ID	75fec42d-ed8f-4fba-a3d0-9ef265dfcdd8	Copy Client ID
-----------	--------------------------------------	----------------

Access level	All projects
Domain allowlist	All

Your Turn! Exercise #1a



1. Create Account/Sign into **Postman**
2. Import Collection & Environment
3. Set username in Environment variables
4. Authenticate with JWT

hint: Authenticate → **Post** Sign in (JWT)

Your Turn! Exercise #1b

1. Find and Run the **Authentication -> Signing In and Out** workflow from the **Sample Workflows** folder

APIs / Sample Workflows / Authentication / **Signing In and Out**



Save



Run

Share



Environment Variables & Field Expressions

Query Workbooks in Site

GET: `{{base-path}}/sites/{{site-id}}/workbooks`

The screenshot shows the Tableau REST API client interface. The left sidebar lists various API endpoints, with 'GET Query Workbooks for Site' highlighted. The main panel shows the selected endpoint with a 'GET' method dropdown and a URL field containing the placeholder `{{base-path}}/sites/{{site-id}}/workbooks`. A 'Send' button is highlighted in the top right. Below the URL bar, there are tabs for 'Params', 'Auth', 'Headers (7)', 'Body', 'Scripts', and 'Settings'. The 'Params' tab is active, displaying a table of parameters:

☐	Key	Value	Description	Bulk Edit
☐	filter	filter-expression	(Optional) An expression that ...	
☐	sort	sort-expression	(Optional) An expression that ...	
☐	pageSize	page-size	(Optional) The number of ite...	
☐	pageNumber	page-number	(Optional) The offset for pagi...	

At the bottom, there is a 'Response' section with a 'History' dropdown and a 'Click Send to get a response' button.

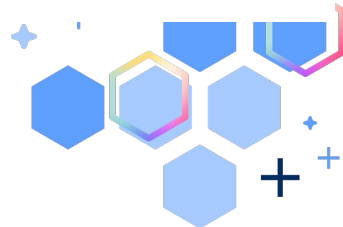
2. Click Send

1. Select Method



What fields can I use in my expression? *hint... a lot.

notes Views
databaseName jobType luid updated At isDefaultPort Users priority hasAlert hasAlerts
tableName parentProjectId progress topLevelProject viewUrlName isPublished sheetType sheetCount
workbookName domainName startedAt isConnectable description hitsTotal friendlyName
size databaseUserName fields
projectId projectName hasEmbeddedPassword createdAt channel name
ownerEmail siteRole ownerName notificationType ownerDomain
title contentUrl updatedAt caption authenticationType favoritesTotal
id connectedWorkbookType workbookDescription tags isHierarchical subscriptionsTotal userId
isLocal hasExtracts type domainNickname displayTabs connectionType completedAt
subtitle serverName status sheetNumber isCertified serverPort completeAt userCount
connectionTo flowId lastLogin



Field Expressions

Return a workbook by name:

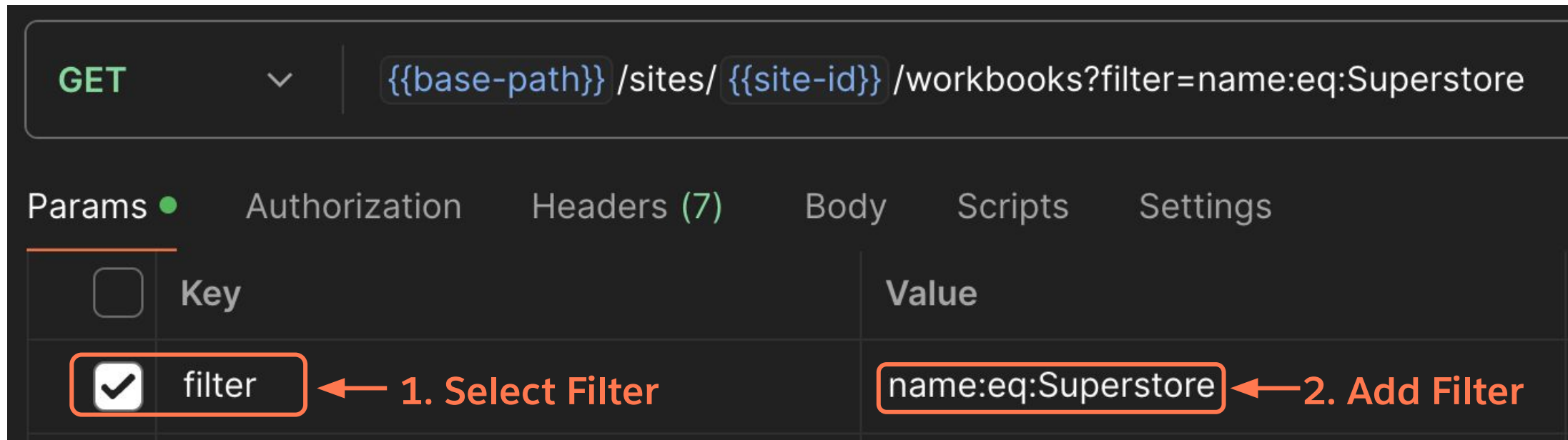
GET: `{{base-path}}/sites/{{site-id}}/workbooks?filter=name:eq:Superstore`

Each expression requires:

Field - “name”

Operator - “eq”

Value - “Superstore”



The screenshot shows a REST client interface with a dark theme. At the top, a dropdown menu is set to 'GET', and the URL is `{{base-path}}/sites/{{site-id}}/workbooks?filter=name:eq:Superstore`. Below the URL bar, there are tabs for 'Params', 'Authorization', 'Headers (7)', 'Body', 'Scripts', and 'Settings'. The 'Params' tab is active, displaying a table with two columns: 'Key' and 'Value'. In the 'Key' column, the first row has a checked checkbox and the text 'filter'. In the 'Value' column, the first row has the text 'name:eq:Superstore'. Red annotations are present: an arrow points from the text '1. Select Filter' to the checked checkbox in the 'filter' row, and another arrow points from the text '2. Add Filter' to the 'name:eq:Superstore' value.

Key	Value
☒ filter	name:eq:Superstore



Set a value as a variable

Select and Right click on the entire **workbook id**

```
Body Cookies Headers (19) Test Results 200 OK • 179 ms • 1.94 KB • Save Response
XML Preview Visualize
1 <?xml version='1.0' encoding='UTF-8'?>
2 <tsResponse xmlns="http://tableau.com/api" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://tableau.com/api https://help.tableau.com/samples/en-us/rest_api/ts-api_3_25.xsd">
3 <pagination pageNumber="1" pageSize="100" totalAvailable="3"/>
4 <workbooks>
5   <workbook id="e4ffb49d-026d-430b-86b6-fde6836b31ca" name="Superstore" contentUrl="Superstore"
    webpageUrl="https://prod-uswest-c.online.tableau.com/#/site/tcrestapihot/workbooks/516006" showTabs="true"
    size="2" createdAt="2025-02-25T20:26:12Z" updatedAt="2025-02-25T20:26:13Z" encryptExtracts="false"
    defaultViewId="1f382bef-20a3-4dce-8216-6778ccf20653">
6     <project id="c76437a3-7572-427f-85f9-1f8d57dfbaa0" name="Samples"/>
7     <location id="c76437a3-7572-427f-85f9-1f8d57dfbaa0" type="Project" name="Samples"/>
8     <owner id="b45ba7fe-a69a-470d-9df8-4af33c219bde" name="jmayo@salesforce.com"/>
9     <tags/>
10    <dataAccelerationConfig/>
11  </workbook>
12  <workbook id="a7d6415d-0686-42e7-a780-71f9cae8c7ff" name="World Indicators" contentUrl="WorldIndicators"
    webpageUrl="https://prod-uswest-c.online.tableau.com/#/site/tcrestapihot/workbooks/516007" showTabs="true"
    size="1" createdAt="2025-02-25T20:26:14Z" updatedAt="2025-02-25T20:26:14Z" encryptExtracts="false"
    defaultViewId="f834d7d0-31ee-4971-8823-17b57a5480b1">
13    <project id="c76437a3-7572-427f-85f9-1f8d57dfbaa0" name="Samples"/>
14    <location id="c76437a3-7572-427f-85f9-1f8d57dfbaa0" type="Project" name="Samples"/>
```

Set a workbook id as a variable

Set the `{{workbook-id}}` variable to this value

The screenshot shows a REST client interface with an XML response. The XML contains a list of workbooks. The first workbook's ID is highlighted, and a context menu is open with 'Set as variable' selected. A search for variables shows 'workbook-id' as the selected option.

1. Highlight and right click

2. Set as variable

3. Select workbook-id

```
<?xml version='1.0' encoding='UTF-8'?>
<tsResponse xmlns="http://tableau.com/api" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://tableau.com/api https://help.tableau.com/samples/en-us/rest_api/ts-api_3_25.xsd">
  <pagination pageNumber="1" pageSize="100" totalAvailable="3"/>
  <workbooks>
    <workbook id="e4ffb49d-026d-430b-86b6-fdc6826b21ee" name="Sample"
      webpageUrl="https://prod-usw...
      size="2" createdAt="2025-02-
      defaultViewId="1f382bef-20a3
    <project id="c76437a3-7572-4
    <location id="c76437a3-7572-
    <owner id="b45ba7fe-a69a-470
    <tags/>
    <dataAccelerationConfig/>
  </workbook>
  <workbook id="a7d6415d-0686-42e7-a780-71f9cae8c7ff" name="World
    webpageUrl="https://prod-uswest-c.online.tableau.com/#/site
    size="1" createdAt="2025-02-25T20:26:14Z" updatedAt="2025-0
    defaultViewId="f834d7d0-31ee-4971-8823-17b57a5480b1">
    <project id="c76437a3-7572-427f-85f9-1f8d57dfbaa0" name="Samples"/>
    <location id="c76437a3-7572-427f-85f9-1f8d57dfbaa0" type="Project" name="Samples"/>
  </workbook>
</tsResponse>
```





Your Turn! Exercise #2

1. First, **Authenticate with JWT**
2. **Query Workbooks for Site, filtering on**
ownerEmail:eq:jmayo@salesforce.com &
name:eq:Superstore
3. Add **Workbook ID** to Environment variable



Requesting View Images, Data, PDF

Query Views for Workbook

Now that the `{{workbook-id}}` is set to Superstore, Query the Views in the Workbook

The screenshot displays the Tableau REST API interface. On the left, a sidebar contains navigation links: Collections, Environments, Flows, History, and a grid icon. The main panel shows a list of API endpoints. The endpoint 'GET Query Views for Workbook' is highlighted with an orange box. The right panel shows the details for this endpoint, including the HTTP method (GET), the URL template, and the query parameters.

Tableau APIs / Tableau REST API / Workbooks and Views / **Query Views for Workbook**

GET `{{base-path}}/sites/{{site-id}}/workbooks/:workbook-id/views`

Params • Authorization Headers (7) Body Scripts Settings

Query Params

☐	Key	Value
☐	includeUsageStatistics	get-usage-information
	Key	Value

Path Variables

	Key	Value
	workbook-id	<code>{{workbook-id}}</code>

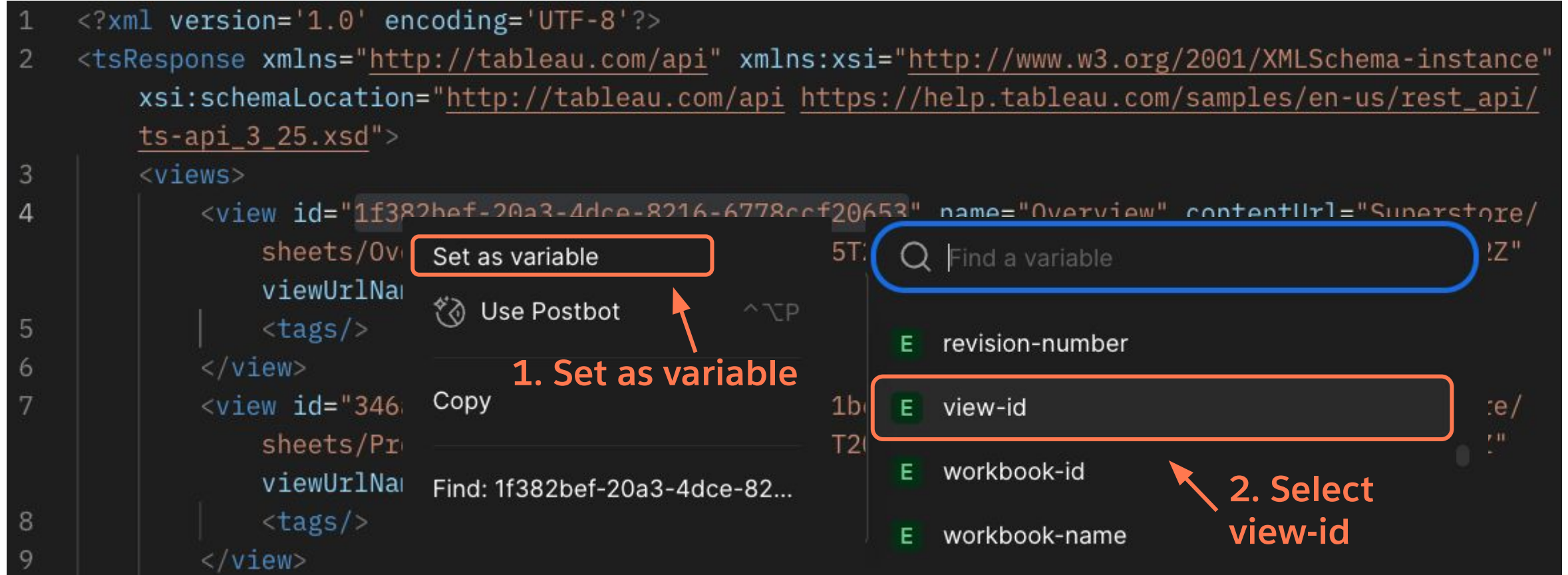
Set view id as a variable

Set the **{{view-id}}** variable to this value

```
1 <?xml version='1.0' encoding='UTF-8'?>
2 <tsResponse xmlns="http://tableau.com/api" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://tableau.com/api https://help.tableau.com/samples/en-us/rest_api/
  ts-api_3_25.xsd">
3   <views>
4     <view id="1f382bef-20a3-4dce-8216-6778ccf20653" name="Overview" contentUrl="Superstore/
      sheets/Overview" viewUrlName="Overview" >
5       <tags/>
6     </view>
7     <view id="346..." name="..." contentUrl="..." viewUrlName="..." >
8       <tags/>
9     </view>
```

1. Set as variable

2. Select view-id

The image shows a code editor with an XML response from the Tableau REST API. The XML contains a list of views. The first view has an id of "1f382bef-20a3-4dce-8216-6778ccf20653". A red box highlights the "Set as variable" button next to this id. An arrow points from the text "1. Set as variable" to this button. To the right, a search bar "Find a variable" is shown with a red box around it. Below the search bar, a list of variables is displayed: "revision-number", "view-id", "workbook-id", and "workbook-name". The "view-id" variable is selected, highlighted with a red box, and an arrow points from the text "2. Select view-id" to it.

Query Views for Workbook

Notice that now **workbook-id** is set to the

The screenshot displays the Tableau REST API interface. On the left sidebar, the 'Query View Data' endpoint is selected and highlighted with an orange border. The main panel shows the configuration for this endpoint:

- Method:** GET
- URL:** `{{base-path}}/sites/{{site-id}}/views/:view-id/data`
- Params:** Authorization, Headers (7), Body, Scripts, Settings
- Query Params:**

Key	Value
maxAge	max-age-minutes
vf_<fieldname>	filter-value
Key	Value

Path Variables:

Key	Value
view-id	{{view-id}}

At the bottom, there are tabs for 'Response' and 'History'.

Query View Image

GET: `{{base-path}}/sites/{{site-id}}/views/:view-id/image?resolution=high`

Tableau APIs / Tableau REST API / Workbooks and Views / Query View Image

GET `{{base-path}}/sites/{{site-id}}/views/:view-id/image?resolution=high`

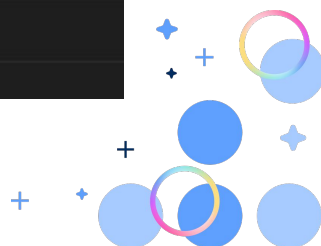
Params • Authorization Headers (7) Body Scripts Settings

Query Params

☐	Key	Value
☒	resolution	high
☐	maxAge	max-age-minutes
☐	vf_<fieldname>	filter-value
	Key	Value

Path Variables

	Key	Value
	view-id	<code>{{view-id}}</code>



Query View Image

GET: `{{base-path}}/sites/{{site-id}}/views/:view-id/image?resolution=high`

GET `{{base-path}}/sites/{{site-id}}/views/:view-id/image?resolution=high` Send

Params • Authorization Headers (7) Body Scripts Settings Cookies

Query Params

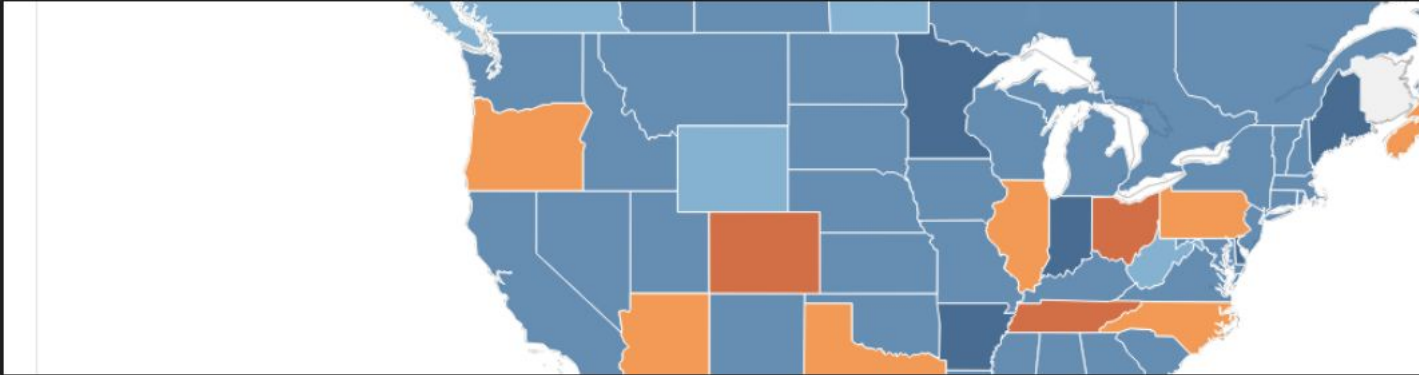
☐ Key	Value	Description	...	Bulk Edit
☒ resolution	high	(Optional) The resolution of the image. Image ...		
☐ maxAge	max-age-minutes	(Optional) The maximum number of minutes a ...		
☐ vf_<fieldname>	filter-value	The value of the field that you want to use to f...		
Key	Value	Description		

Path Variables

Key	Value	Description	...	Bulk Edit
view-id	<code>{{view-id}}</code>	Description		

Body Cookies Headers (17) Test Results 200 OK • 5.55 s • 398.32 KB Save Response

Hex Preview Visualize

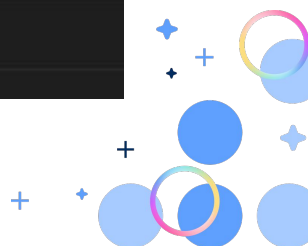


Query View PDF

GET: `{{base-path}}/sites/{{site-id}}/views/:view-id/pdf?orientation=Landscape`

The screenshot displays the Tableau REST API Explorer interface. On the left sidebar, the 'GET Query View PDF' endpoint is selected and highlighted with an orange box. The main panel shows the endpoint URL: `{{base-path}}/sites/{{site-id}}/views/:view-id/pdf?orientation=Landscape`, with the `orientation=Landscape` part underlined in orange. Below the URL, the 'Params' tab is active, showing a list of parameters. The 'orientation' parameter is checked and highlighted with an orange box, with its value set to 'Landscape'.

Key	Value
☐ vizHeight	viz-height
☐ vizWidth	viz-width
☐ maxAge	max-age-minutes
☐ type	page-type
☒ orientation	Landscape
☐ vf_<fieldname>	filter-value
Key	Value



Query View PDF

GET: `{{base-path}}/sites/{{site-id}}/views/:view-id/pdf?orientation=Landscape`

GET `{{base-path}}/sites/{{site-id}}/views/:view-id/pdf?orientation=Landscape` Send

Params • Authorization Headers (7) Body Scripts Settings Cookies

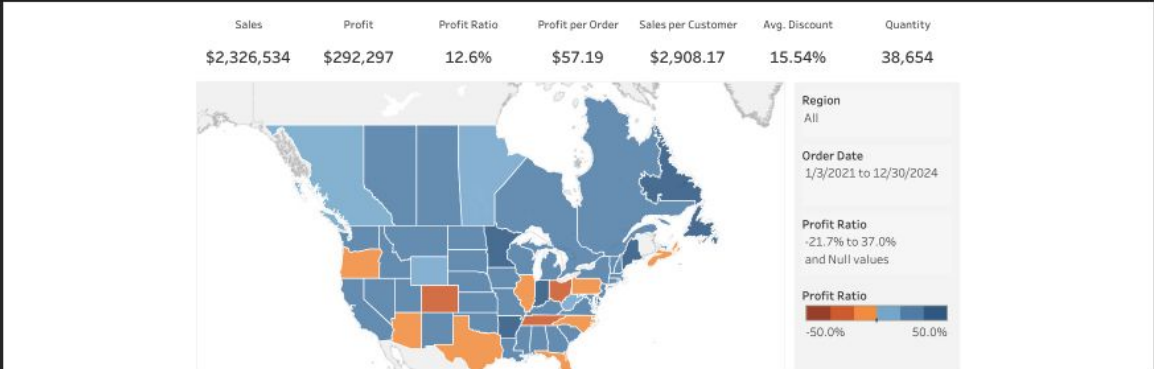
☐ Key	Value	Description	...	Bulk Edit
☐ vizHeight	viz-height	(Optional) The height of the rendered pdf ima...		
☐ vizWidth	viz-width	(Optional) The width of the rendered pdf ima...		
☐ maxAge	max-age-minutes	(Optional) The maximum number of minutes ...		
☐ type	page-type	(Optional) The type of page, which determine...		
☒ orientation	Landscape	(Optional) The orientation of the pages in the...		
☐ vf_<fieldname>	filter-value	The value of the field that you want to use to ...		
☐ Key	Value	Description		

Path Variables

Key	Value	Description	...	Bulk Edit
-----	-------	-------------	-----	-----------

Body Cookies Headers (17) Test Results 200 OK • 3.99 s • 171.72 KB • Save Response

Hex Preview Visualize



Sales: \$2,326,534 Profit: \$292,297 Profit Ratio: 12.6% Profit per Order: \$57.19 Sales per Customer: \$2,908.17 Avg. Discount: 15.54% Quantity: 38,654

Region: All

Order Date: 1/3/2021 to 12/30/2024

Profit Ratio: -21.7% to 37.0% and Null values

Profit Ratio: -50.0% to 50.0%



Query View Preview Image - Thumbnail

GET:{{base-path}}/sites/{{site-id}}/workbooks/:workbook-id/views/:view-id/
previewImage

The screenshot displays the Tableau REST API Explorer interface. On the left sidebar, the 'History' section is active, showing a list of API endpoints. The endpoint 'GET Query View Preview Image' is highlighted with an orange border. The main panel on the right shows the details for this endpoint, including the HTTP method (GET), the full URL with path variables, and tabs for Params, Authorization, Headers, Body, Scripts, and Settings. The 'Params' tab is selected, showing 'Query Params' and 'Path Variables'.

Tableau APIs / Tableau REST API / Workbooks and Views / Query View Preview Image

GET `{{base-path}}/sites/{{site-id}}/workbooks/:workbook-id/views/:view-id/previewImage`

Params • Authorization Headers (7) Body Scripts Settings

Query Params

	Key	Value
	Key	Value

Path Variables

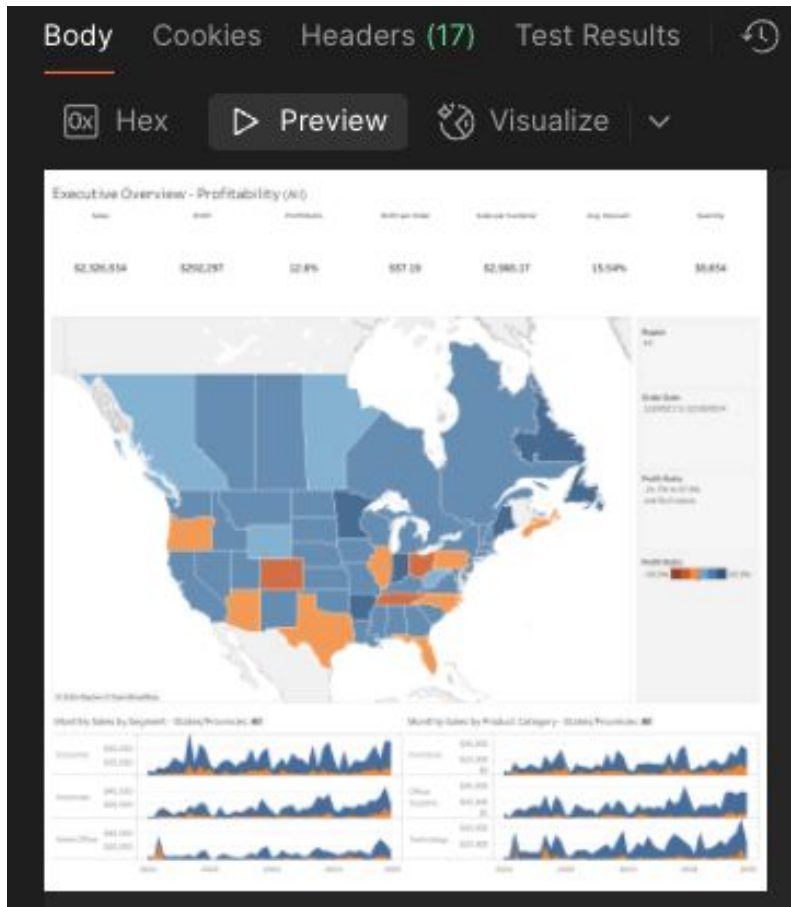
	Key	Value
	workbook-id	<code>{{workbook-id}}</code>
	view-id	<code>{{view-id}}</code>

Query View Preview Image - Thumbnail

GET:{{base-path}}/sites/{{site-id}}/workbooks/:workbook-id/views/:view-id/
previewImage

Permissions:

Tableau users who are not administrators can query workbook views only if they have Read (view) permission for the views (either explicitly or implicitly).



Body 403 Forbidden 349 ms 1.58 KB Save Response

```
<?xml version='1.0' encoding='UTF-8'?>
<tsResponse xmlns="http://tableau.com/api" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="http://tableau.com/api https://help.tableau.com/samples/en-us/rest_api/ts-api_3_25.xsd">
  <error code="403004">
    <summary>Forbidden</summary>
    <detail>'user311@tc25.com' isn't authorized to query the preview image for view '3c75de28-5d76-4e8b-b924-5fec50a02dc1'.</detail>
  </error>
</tsResponse>
```

Query View Data

GET: `{{base-path}}/sites/{{site-id}}/views/:view-id/data`

The screenshot displays the Tableau REST API Explorer interface. On the left sidebar, the 'Environments' section is active, and the 'Query View Data' endpoint is selected and highlighted with an orange border. The main panel shows the details for this endpoint, including the HTTP method (GET), the URL template, and the query parameters.

Tableau APIs / Tableau REST API / Workbooks and Views / **Query View Data**

GET `{{base-path}}/sites/{{site-id}}/views/:view-id/data`

Params • Authorization Headers (7) Body Scripts Settings

Query Params

☐	Key	Value
☐	maxAge	max-age-minutes
☐	vf_<fieldname>	filter-value
	Key	Value

Path Variables

	Key	Value
	view-id	<code>{{view-id}}</code>

Query View Data

GET: `{{base-path}}/sites/{{site-id}}/views/:view-id/data`

```
Body 200 OK • 3.15 s • 3.47 KB | Save Response
```

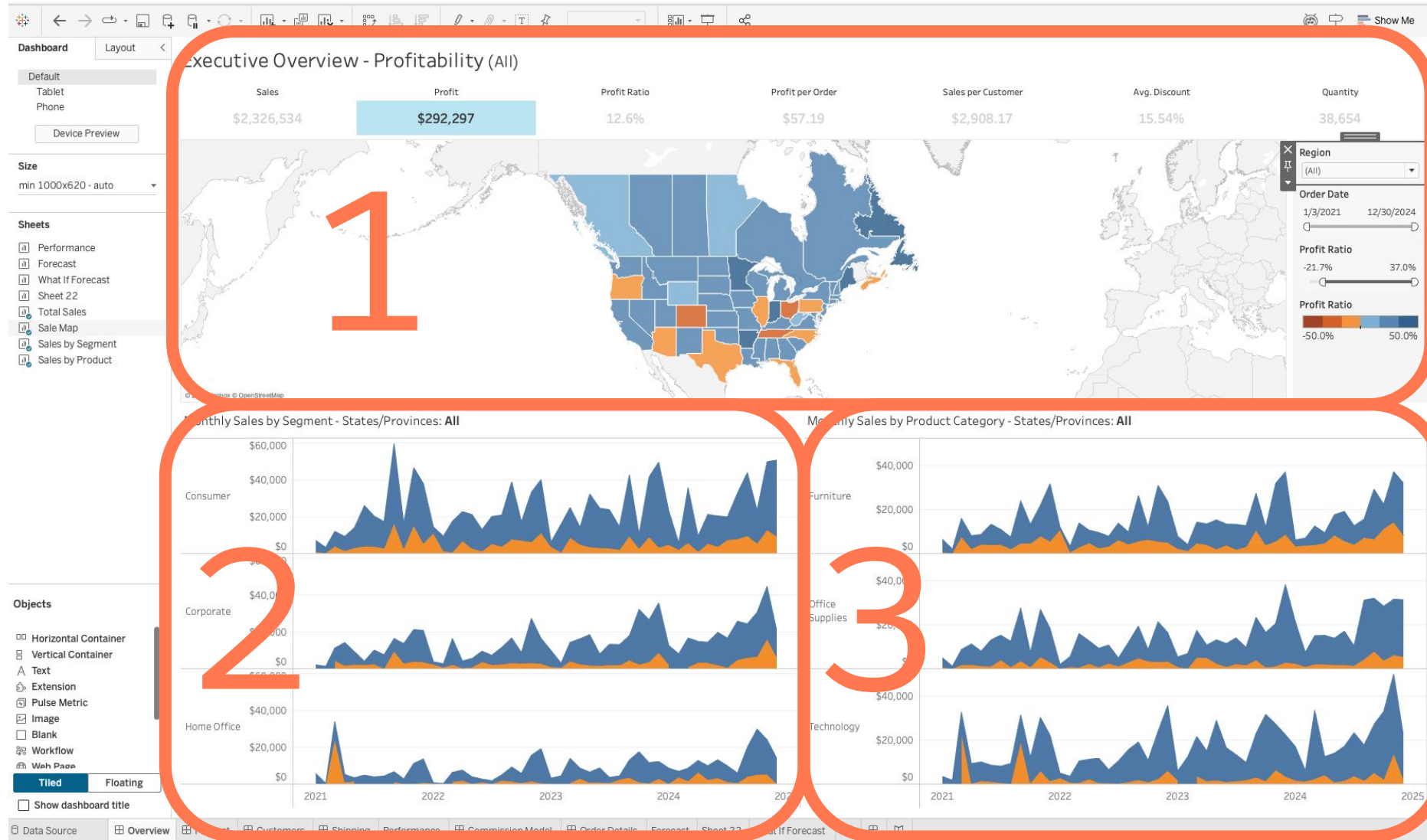
Raw Preview Visualize

```
1 Country/Region,State/Province,Profit Ratio,Latitude (generated),Longitude
   (generated)
2 Canada,Alberta,19.5%,53.41,-114.42
3 Canada,British Columbia,4.2%,54.9464,-125.1024
4 Canada,Manitoba,8.2%,55.0085,-97.1771
5 Canada,Newfoundland and Labrador,36.2%,53.63,-61.37
6 Canada,Nova Scotia,-13.8%,45,-63.58
7 Canada,Ontario,26.8%,50.94,-84.75
8 Canada,Prince Edward Island,30.3%,46.32,-63.19
9 Canada,Quebec,28.5%,53.02,-72.58
10 Canada,Saskatchewan,21.6%,54.49,-105.68
11 United States,Alabama,29.7%,32.75,-86.75
12 United States,Arizona,-9.7%,34.5,-111.5
13 United States,Arkansas,34.3%,34.75,-92.5
14 United States,California,16.7%,37.25,-119.75
15 United States,Colorado,-20.3%,39,-105.5
16 United States,Connecticut,26.2%,41.6227,-72.7028
17 United States,Delaware,36.3%,38.6716,-75.4165
18 United States,District of Columbia,37.0%,38.91,-77
19 United States,Florida,-3.8%,28.42,-81.55
20 United States,Georgia,33.1%,32.1598,-83.4715
21 United States,Idaho,18.9%,44.0739,-114.542
22 United States,Illinois,-15.7%,40,-89.25
23 United States,Indiana,34.3%,40,-86.25
24 United States,Iowa,25.8%,42,-93.5
```



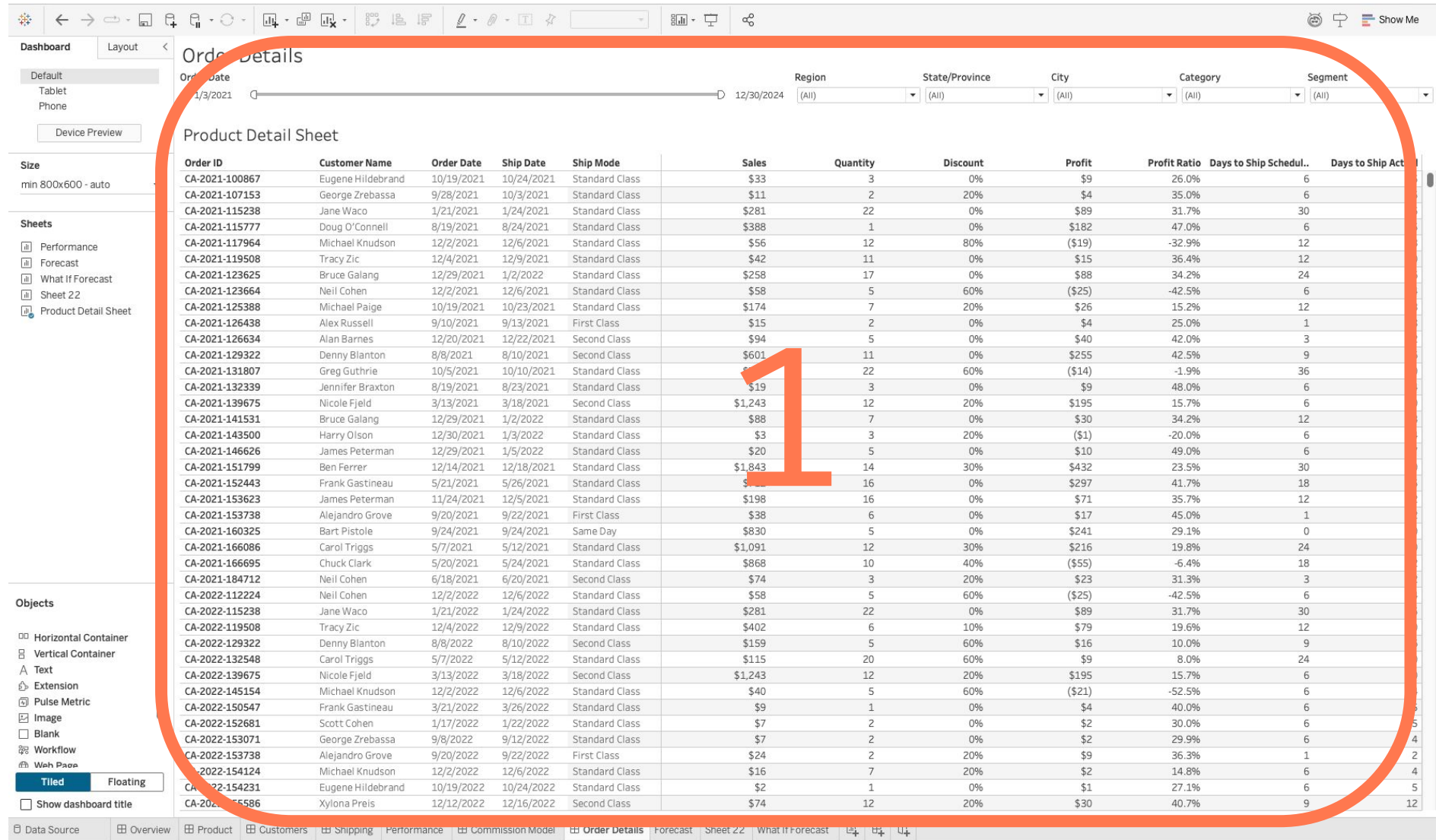
Query View Data - Only pulls data from first View

GET: `{{base-path}}/sites/{{site-id}}/views/:view-id/data`



Query View Data - Tabular View

GET: `{{base-path}}/sites/{{site-id}}/views/:view-id/data`



Order Details

Product Detail Sheet

Order ID	Customer Name	Order Date	Ship Date	Ship Mode	Sales	Quantity	Discount	Profit	Profit Ratio	Days to Ship Schedul..	Days to Ship Act
CA-2021-100867	Eugene Hildebrand	10/19/2021	10/24/2021	Standard Class	\$33	3	0%	\$9	26.0%	6	
CA-2021-107153	George Zrebassa	9/28/2021	10/3/2021	Standard Class	\$11	2	20%	\$4	35.0%	6	
CA-2021-115238	Jane Waco	1/21/2021	1/24/2021	Standard Class	\$281	22	0%	\$89	31.7%	30	
CA-2021-115777	Doug O'Connell	8/19/2021	8/24/2021	Standard Class	\$388	1	0%	\$182	47.0%	6	
CA-2021-117964	Michael Knudson	12/2/2021	12/6/2021	Standard Class	\$56	12	80%	(\$19)	-32.9%	12	
CA-2021-119508	Tracy Zic	12/4/2021	12/9/2021	Standard Class	\$42	11	0%	\$15	36.4%	12	
CA-2021-123625	Bruce Galang	12/29/2021	1/2/2022	Standard Class	\$258	17	0%	\$88	34.2%	24	
CA-2021-123664	Neil Cohen	12/2/2021	12/6/2021	Standard Class	\$58	5	60%	(\$25)	-42.5%	6	
CA-2021-125388	Michael Paige	10/19/2021	10/23/2021	Standard Class	\$174	7	20%	\$26	15.2%	12	
CA-2021-126438	Alex Russell	9/10/2021	9/13/2021	First Class	\$15	2	0%	\$4	25.0%	1	
CA-2021-126634	Alan Barnes	12/20/2021	12/22/2021	Second Class	\$94	5	0%	\$40	42.0%	3	
CA-2021-129322	Denny Blanton	8/8/2021	8/10/2021	Second Class	\$601	11	0%	\$255	42.5%	9	
CA-2021-131807	Greg Guthrie	10/5/2021	10/10/2021	Standard Class	\$1	22	60%	(\$14)	-1.9%	36	
CA-2021-132339	Jennifer Braxton	8/19/2021	8/23/2021	Standard Class	\$19	3	0%	\$9	48.0%	6	
CA-2021-139675	Nicole Fjeld	3/13/2021	3/18/2021	Second Class	\$1,243	12	20%	\$195	15.7%	6	
CA-2021-141531	Bruce Galang	12/29/2021	1/2/2022	Standard Class	\$88	7	0%	\$30	34.2%	12	
CA-2021-143500	Harry Olson	12/30/2021	1/3/2022	Standard Class	\$3	3	20%	(\$1)	-20.0%	6	
CA-2021-146626	James Peterman	12/29/2021	1/5/2022	Standard Class	\$20	5	0%	\$10	49.0%	6	
CA-2021-151799	Ben Ferrer	12/14/2021	12/18/2021	Standard Class	\$1,843	14	30%	\$432	23.5%	30	
CA-2021-152443	Frank Gastineau	5/21/2021	5/26/2021	Standard Class	\$1	16	0%	\$297	41.7%	18	
CA-2021-153623	James Peterman	11/24/2021	12/5/2021	Standard Class	\$198	16	0%	\$71	35.7%	12	
CA-2021-153738	Alejandro Grove	9/20/2021	9/22/2021	First Class	\$38	6	0%	\$17	45.0%	1	
CA-2021-160325	Bart Pistole	9/24/2021	9/24/2021	Same Day	\$830	5	0%	\$241	29.1%	0	
CA-2021-166086	Carol Triggs	5/7/2021	5/12/2021	Standard Class	\$1,091	12	30%	\$216	19.8%	24	
CA-2021-166695	Chuck Clark	5/20/2021	5/24/2021	Standard Class	\$868	10	40%	(\$55)	-6.4%	18	
CA-2021-184712	Neil Cohen	6/18/2021	6/20/2021	Second Class	\$74	3	20%	\$23	31.3%	3	
CA-2022-112224	Neil Cohen	12/2/2022	12/6/2022	Standard Class	\$58	5	60%	(\$25)	-42.5%	6	
CA-2022-115238	Jane Waco	1/21/2022	1/24/2022	Standard Class	\$281	22	0%	\$89	31.7%	30	
CA-2022-119508	Tracy Zic	12/4/2022	12/9/2022	Standard Class	\$402	6	10%	\$79	19.6%	12	
CA-2022-129322	Denny Blanton	8/8/2022	8/10/2022	Second Class	\$159	5	60%	\$16	10.0%	9	
CA-2022-132548	Carol Triggs	5/7/2022	5/12/2022	Standard Class	\$115	20	60%	\$9	8.0%	24	
CA-2022-139675	Nicole Fjeld	3/13/2022	3/18/2022	Second Class	\$1,243	12	20%	\$195	15.7%	6	
CA-2022-145154	Michael Knudson	12/2/2022	12/6/2022	Standard Class	\$40	5	60%	(\$21)	-52.5%	6	
CA-2022-150547	Frank Gastineau	3/21/2022	3/26/2022	Standard Class	\$9	1	0%	\$4	40.0%	6	
CA-2022-152681	Scott Cohen	1/17/2022	1/22/2022	Standard Class	\$7	2	0%	\$2	30.0%	6	
CA-2022-153071	George Zrebassa	9/8/2022	9/12/2022	Standard Class	\$7	2	0%	\$2	29.9%	6	
CA-2022-153738	Alejandro Grove	9/20/2022	9/22/2022	First Class	\$24	2	20%	\$9	36.3%	1	
CA-2022-154124	Michael Knudson	12/2/2022	12/6/2022	Standard Class	\$16	7	20%	\$2	14.8%	6	
CA-2022-154231	Eugene Hildebrand	10/19/2022	10/24/2022	Standard Class	\$2	1	0%	\$1	27.1%	6	
CA-2022-155586	Xylona Preis	12/12/2022	12/16/2022	Second Class	\$74	12	20%	\$30	40.7%	9	

Query View Data - Tabular View

GET: `{{base-path}}/sites/{{site-id}}/views/:view-id/data`

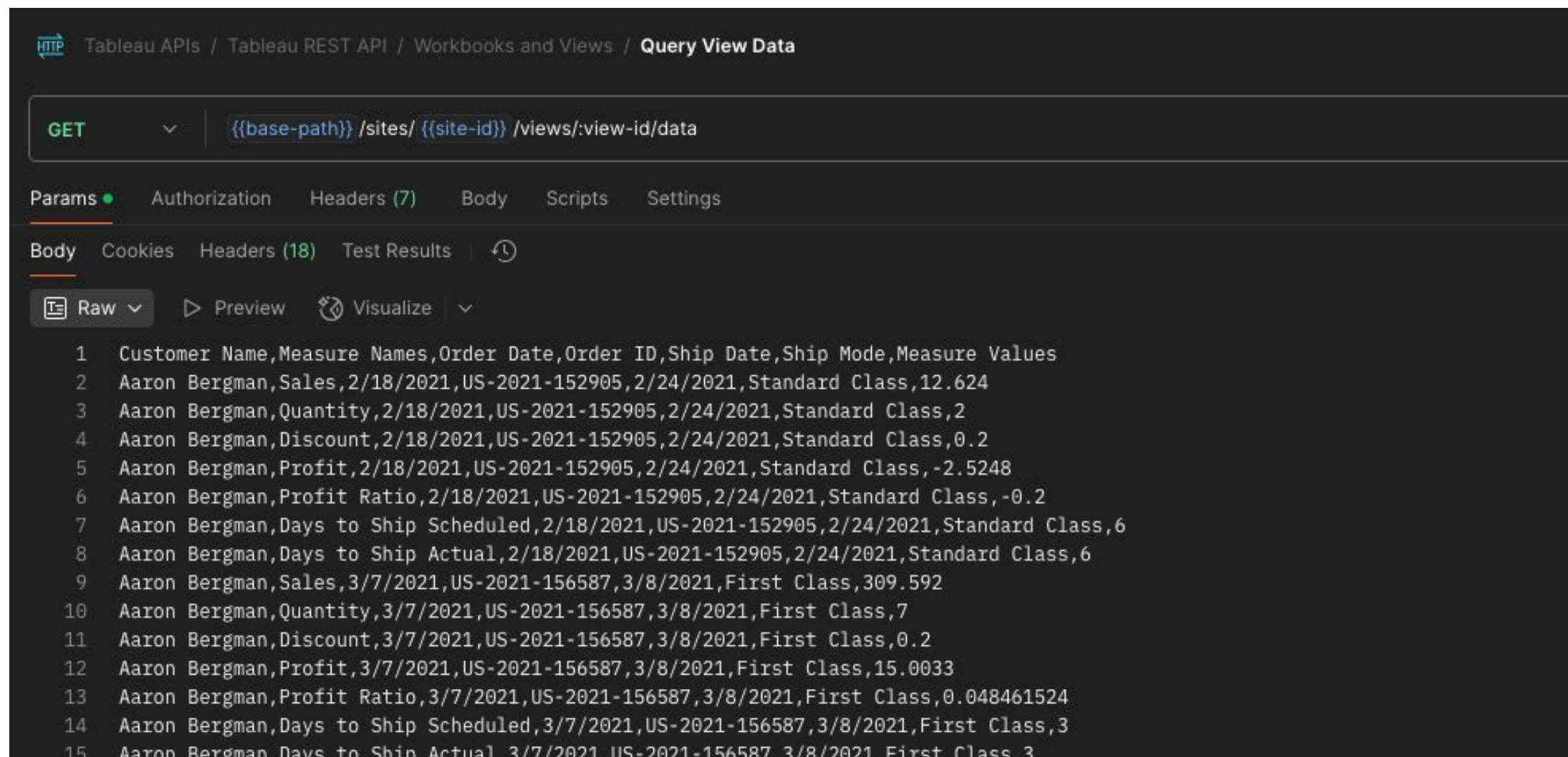


Tableau APIs / Tableau REST API / Workbooks and Views / Query View Data

GET `{{base-path}}/sites/{{site-id}}/views/:view-id/data`

Params • Authorization Headers (7) Body Scripts Settings

Body Cookies Headers (18) Test Results ↻

Raw ▾ Preview Visualize ▾

1	Customer Name,Measure Names,Order Date,Order ID,Ship Date,Ship Mode,Measure Values
2	Aaron Bergman,Sales,2/18/2021,US-2021-152905,2/24/2021,Standard Class,12.624
3	Aaron Bergman,Quantity,2/18/2021,US-2021-152905,2/24/2021,Standard Class,2
4	Aaron Bergman,Discount,2/18/2021,US-2021-152905,2/24/2021,Standard Class,0.2
5	Aaron Bergman,Profit,2/18/2021,US-2021-152905,2/24/2021,Standard Class,-2.5248
6	Aaron Bergman,Profit Ratio,2/18/2021,US-2021-152905,2/24/2021,Standard Class,-0.2
7	Aaron Bergman,Days to Ship Scheduled,2/18/2021,US-2021-152905,2/24/2021,Standard Class,6
8	Aaron Bergman,Days to Ship Actual,2/18/2021,US-2021-152905,2/24/2021,Standard Class,6
9	Aaron Bergman,Sales,3/7/2021,US-2021-156587,3/8/2021,First Class,309.592
10	Aaron Bergman,Quantity,3/7/2021,US-2021-156587,3/8/2021,First Class,7
11	Aaron Bergman,Discount,3/7/2021,US-2021-156587,3/8/2021,First Class,0.2
12	Aaron Bergman,Profit,3/7/2021,US-2021-156587,3/8/2021,First Class,15.0033
13	Aaron Bergman,Profit Ratio,3/7/2021,US-2021-156587,3/8/2021,First Class,0.048461524
14	Aaron Bergman,Days to Ship Scheduled,3/7/2021,US-2021-156587,3/8/2021,First Class,3
15	Aaron Bergman,Days to Ship Actual,3/7/2021,US-2021-156587,3/8/2021,First Class,3

Your Turn! Exercise #3



1. **Query Views for Workbook** (uses Superstore)
2. Set the view ID for a view of your choice as the **{{view-id}}** variable
3. **Query View Image** and click “Send”
4. **Query View PDF** and click “Send”
5. **Query View Data** and click “Send”



Data & Filter Expressions

Query View Data

GET: `{{base-path}}/sites/{{site-id}}/views/:view-id/data`

Body 200 OK 3.15 s 3.47 KB Save Response

Raw Preview Visualize

1	Country/Region,State/Province,Profit Ratio,Latitude (generated),Longitude (generated)
2	Canada,Alberta,19.5%,53.41,-114.42
3	Canada,British Columbia,4.2%,54.9464,-125.1024
4	Canada,Manitoba,8.2%,55.0085,-97.1771
5	Canada,Newfoundland and Labrador,36.2%,53.63,-61.37
6	Canada,Nova Scotia,-13.8%,45,-63.58
7	Canada,Ontario,26.8%,50.94,-84.75
8	Canada,Prince Edward Island,30.3%,46.32,-63.19
9	Canada,Quebec,28.5%,53.02,-72.58
10	Canada,Saskatchewan,21.6%,54.49,-105.68
11	United States,Alabama,29.7%,32.75,-86.75
12	United States,Arizona,-9.7%,34.5,-111.5
13	United States,Arkansas,34.3%,34.75,-92.5
14	United States,California,16.7%,37.25,-119.75
15	United States,Colorado,-20.3%,39,-105.5
16	United States,Connecticut,26.2%,41.6227,-72.7028
17	United States,Delaware,36.3%,38.6716,-75.4165
18	United States,District of Columbia,37.0%,38.91,-77
19	United States,Florida,-3.8%,28.42,-81.55
20	United States,Georgia,33.1%,32.1598,-83.4715
21	United States,Idaho,18.9%,44.0739,-114.542
22	United States,Illinois,-15.7%,40,-89.25
23	United States,Indiana,34.3%,40,-86.25
24	United States,Iowa,25.8%,42,-93.5

Body 200 OK 3.14 s 3.47 KB Save Response

Raw Preview Visualize

36	United States,Nebraska,27.3%,41.5,-99.75
37	United States,Nevada,19.8%,39.25,-116.75
38	United States,New Hampshire,23.4%,43.66,-71.49
39	United States,New Jersey,27.3%,40.16,-74.49
40	United States,New Mexico,24.2%,34.5,-106
41	United States,New York,23.8%,43,-75.49
42	United States,North Carolina,-13.5%,35.5,-80
43	United States,North Dakota,25.0%,47.5,-100
44	United States,Ohio,-21.7%,40.25,-83
45	United States,Oklahoma,24.7%,35.5,-97.5
46	United States,Oregon,-6.8%,44,-120.5
47	United States,Pennsylvania,-13.4%,40.86,-77.9
48	United States,Rhode Island,32.2%,41.4966,-71.4757
49	United States,South Carolina,20.9%,34,-81
50	United States,South Dakota,30.0%,44.5,-100.25
51	United States,Tennessee,-17.4%,35.75,-86.25
52	United States,Texas,-15.1%,31.25,-99.25
53	United States,Utah,22.7%,39.25,-111.75
54	United States,Vermont,25.1%,44.1584,-72.7678
55	United States,Virginia,26.3%,37.77,-78.24
56	United States,Washington,24.1%,47.5,-120.5
57	United States,West Virginia,15.4%,38.63,-80.73
58	United States,Wisconsin,26.2%,44.5,-90
59	United States,Wyoming,6.3%,43,-107.5
60	

59 Results

Query View Data - with Filter Expression

GET:

The screenshot displays the Tableau REST API interface for the 'Query View Data' endpoint. The left sidebar shows a list of API endpoints, with 'GET Query View Data' highlighted. The main panel shows the endpoint URL: `{{base-path}}/sites/{{site-id}}/views/:view-id/data?vf_Region=West`. Below the URL, the 'Params' tab is selected, showing a table of query parameters. The 'vf_Region' parameter is checked and set to 'West'. The 'Path Variables' section shows the 'view-id' variable set to `{{view-id}}`.

Tableau APIs / Tableau REST API / Workbooks and Views / Query View Data

GET `{{base-path}}/sites/{{site-id}}/views/:view-id/data?vf_Region=West`

Params • Authorization Headers (7) Body Scripts Settings

Query Params

☐	Key	Value
☐	maxAge	max-age-minutes
☒	vf_Region	West
	Key	Value

Path Variables

	Key	Value
	view-id	<code>{{view-id}}</code>

Query View Data

GET: `{{base-path}}/sites/{{site-id}}/views/:view-id/data?vf_Region=West`

Body 200 OK • 2.65 s • 1.63 KB • Save Response

Raw Preview Visualize

	Country/Region,State/Province,Profit Ratio,Latitude (generated),Longitude (generated)
1	
2	Canada,Alberta,19.5%,53.41,-114.42
3	Canada,British Columbia,4.2%,54.9464,-125.1024
4	Canada,Saskatchewan,21.6%,54.49,-105.68
5	United States,Arizona,-9.7%,34.5,-111.5
6	United States,California,16.7%,37.25,-119.75
7	United States,Colorado,-20.3%,39,-105.5
8	United States,Idaho,18.9%,44.0739,-114.542
9	United States,Montana,32.8%,47,-109.75
10	United States,Nevada,19.8%,39.25,-116.75
11	United States,New Mexico,24.2%,34.5,-106
12	United States,Oregon,-6.8%,44,-120.5
13	United States,Utah,22.7%,39.25,-111.75
14	United States,Washington,24.1%,47.5,-120.5
15	United States,Wyoming,6.3%,43,-107.5
16	

← Only West Region Results



Your Turn! Exercise #4



1. **Query View Data** for **“Order Details”** View
2. Check the **Params** box for **vf_<fieldname>** and set it to **vf_Ship Mode** with a value of **“First Class”**
3. Click **“Send”** to see the results
(How many First Class shipments are there?)



Headless BI using the *NEW* VizQL Data Service API

Tableau is a Full-Stack Analytics Platform

Not just for visualization!

Use cases – App Integration, Business Process Integration, Tableau Embedding, Data Science, GenAI, and more

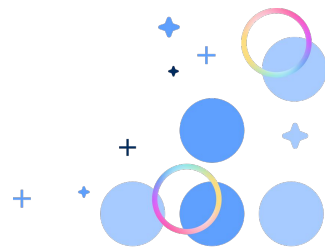
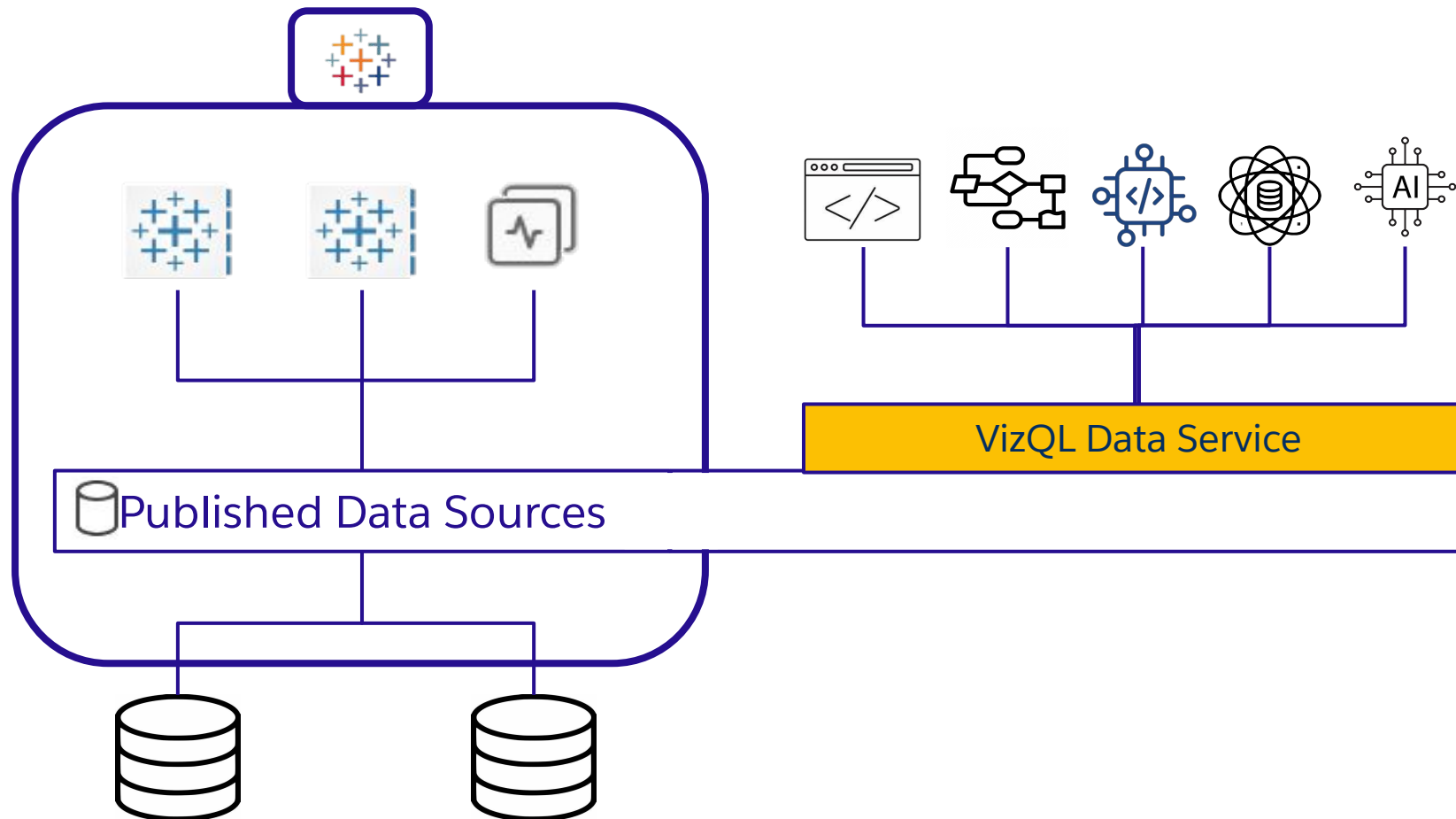
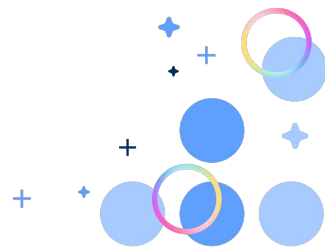
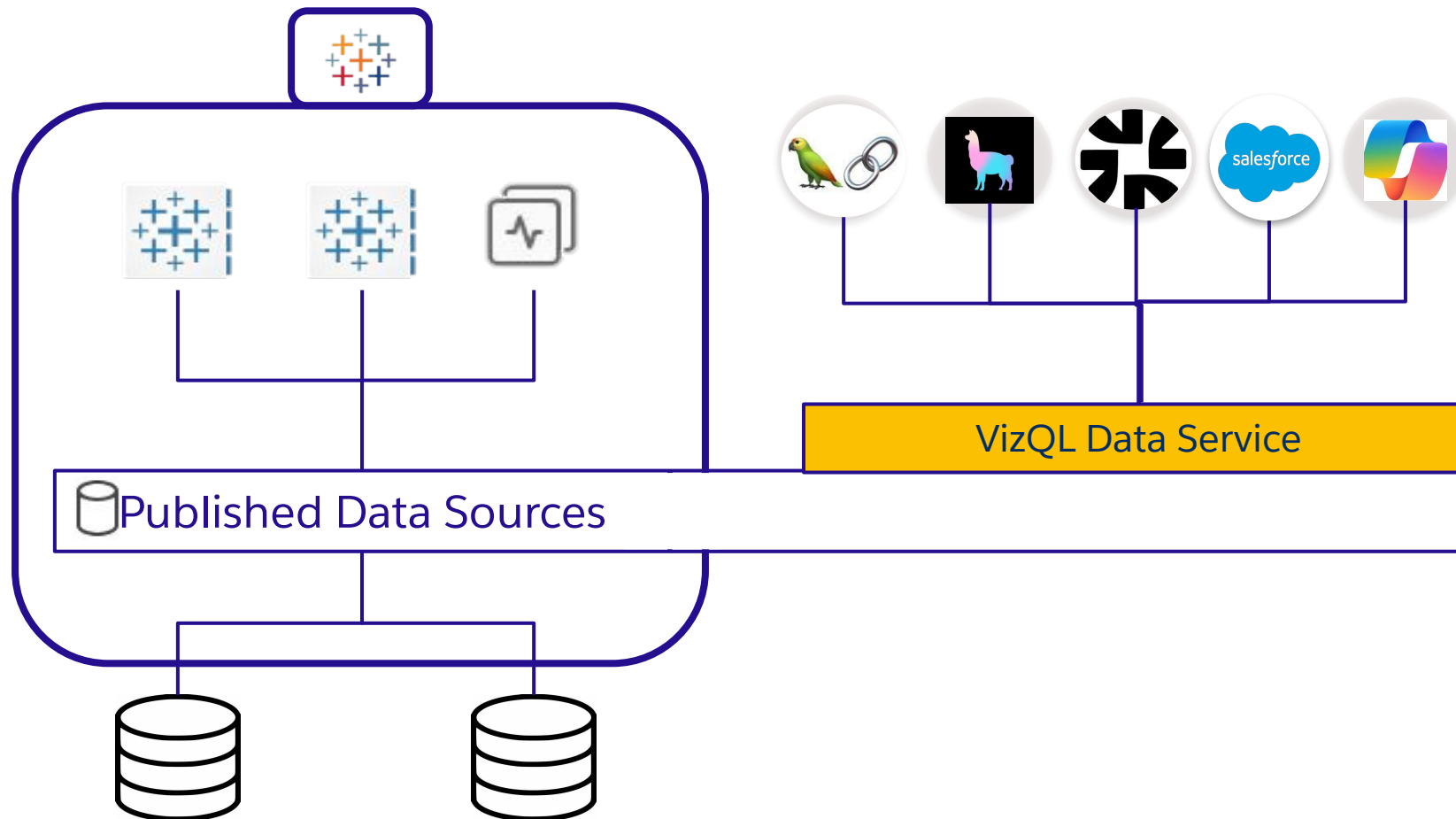


Tableau is a Full-Stack Analytics Platform

Not just for visualization!

Use cases – App Integration, Business Process Integration, Tableau Embedding, Data Science, GenAI, and more



What is VizQL Data Service?

- A new Tableau service that sits alongside our existing REST services
- Accessed via HTTP and uses JSON as the content type
- Uses standard Tableau Authentication
- Two primary endpoints
 - **Read Metadata** - to get information about a Published Datasource
 - **Query Datasource** - to send a query to a Published Datasource
- Provides a simple, well defined API for Query Semantics
- Available on Tableau Server and Tableau Cloud in 2025.1

VizQL Data Service API

Tableau APIs / VizQL Data Service

Overview Authorization Scripts

VizQL Data Service

VizQL Data Service Collection

The VizQL Data Service provides a programmatic way for you to access data from your published data sources. Use this collection to retrieve data and field metadata from a specific datasource.

See full documentation here: [VizQL Data Service](#)

Get Started

To get started, see [How to Use the Collection](#).

Authentication

For authentication, see [Tableau REST API > Authenticate](#).

Find your datasource luid

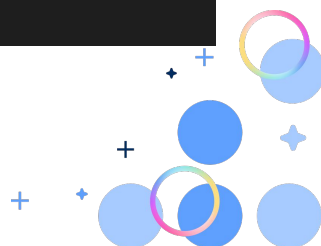
- > Mobile Settings
- > Notifications
- > Permissions
- > Projects
- > Publishing
- > Pulse (Cloud-only)
- > Revisions
- > Server
- > Sites
- > Subscriptions
- > Users and Groups
- > Virtual Connections
- > Workbooks and Views
- > Pulse Test
- > Tableau Metadata API (Grap...
- ✓ **VizQL Data Service**
 - POST Read Metadata
 - > POST Query Datasource

Query Data Sources

GET: `{{base-path}}/sites/{{site-id}}/datasources`

The screenshot displays the Tableau REST API interface. On the left sidebar, the 'Data Sources' collection is expanded, and the 'GET Query Data Sources' endpoint is highlighted with an orange border. The main panel shows the details for this endpoint, including the HTTP method (GET) and the URL template `{{base-path}}/sites/{{site-id}}/datasources`. Below the URL, the 'Params' tab is active, showing a table of query parameters.

Key	Value
pageSize	100
pageNumber	1
filter	
sort	
fields	
Key	Value



Set a datasource id as a variable

Set the `{{datasource-id}}` variable to this value

The screenshot shows a web browser displaying an XML response from a Tableau API. The XML structure is as follows:

```
1 <?xml version='1.0' encoding='UTF-8'?>
2 <response xmlns="http://tableau.com/api" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="http://tableau.com/api https://help.tableau.com/samples/en-
3 <pagination pageNumber="1" pageSize="100" totalAvailable="10"/>
4 <datasources>
5   <datasource contentUrl="SuperstoreDatasource" createdAt="2025-02-25T20:26:15Z" encryptExtracts="false" hasExtracts="false" id="95edde97-c967-419d-98ad-0ba4fb47a997" isC
6     <project id="c76437a3-7572-427f-85f9-1f8d57dfbaa0" name="Samples"/>
7     <owner id="b45ba7fe-a69a-470d-9df8-4af33c219bde" name="jmayo@salesforce.com"/>
8     <tags/>
9   </datasource>
10  <datasource contentUrl="TSUsers" createdAt="2025-02-26T17:31:31Z" description="*Ov
11    <project id="8bdbfa2b-1846-4e5b-818e-bba744b73a4a" name="Admin Insights"/>
12    <owner id="6a615a5a-8789-4042-ae62-0e4e4124b084" name="tol.admin.api.broker.se
13    <tags/>
14  </datasource>
15  <datasource contentUrl="Permissions" createdAt="2025-02-26T17:31:32Z" description=
16    <project id="8bdbfa2b-1846-4e5b-818e-bba744b73a4a" name="Admin Insights"/>
17    <owner id="6a615a5a-8789-4042-ae62-0e4e4124b084" name="tol.admin.api.broker.se
18    <tags/>
19  </datasource>
20  <datasource contentUrl="TSEvents" createdAt="2025-02-26T17:31:32Z" description="*O
21    <project id="8bdbfa2b-1846-4e5b-818e-bba744b73a4a" name="Admin Insights"/>
22    <owner id="6a615a5a-8789-4042-ae62-0e4e4124b084" name="tol.admin.api.broker.service.usera@tableau.com"/>
23  </datasource>
```

A search bar is used to find a variable in the XML. The search results show the following variables:

- analytics-extensions-connection-luid
- datasource-name
- datasource-id** (selected)
- flow-id
- flow-name
- task-id

The context menu for the selected variable shows the following options:

- Set as variable
- Use Postbo
- Copy
- Find: 95edde97-c967-419d-9...

Annotations on the image:

1. Set as variable
2. Select datasource-id

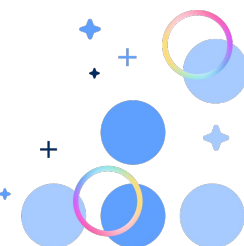
Read Metadata

POST: `{{vizql-base-path}}/read-metadata`

The screenshot displays a REST client interface with the following details:

- URL:** `{{vizql-base-path}}/read-metadata`
- Method:** POST
- Body:** JSON
- Response:** A JSON object with a `"data"` array containing two field objects.

```
{
  "data": [
    {
      "fieldName": "Calculation_1368249927221915648",
      "fieldCaption": "Profit Ratio",
      "dataType": "REAL",
      "logicalTableId": ""
    },
    {
      "fieldName": "Category",
      "fieldCaption": "Category",
      "dataType": "STRING",
      "logicalTableId": "Orders_ECFC1FB690A41FE803BC071773BA862"
    }
  ]
}
```



Query Datasource simple-query

POST: `{{vizql-base-path}}/query-datasource`

Tableau APIs / VizQL Data Service / Query Datasource • From `simple-query`

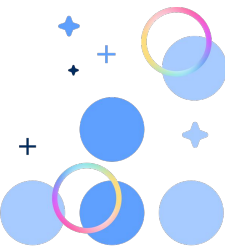
POST `{{vizql-base-path}}/query-datasource`

Params Authorization Headers (9) Body • Scripts Settings

Body Cookies (1) Headers (13) Test Results

`{}` JSON Preview Visualize

```
1 {
2   "data": [
3     {
4       "Category": "Technology",
5       "SUM(Sales)": 839893.2789999967
6     },
7     {
8       "Category": "Furniture",
9       "SUM(Sales)": 754747.7612999984
10    },
11    {
12      "Category": "Office Supplies",
13      "SUM(Sales)": 731893.3140000021
14    }
15  ]
16 }
```





Your Turn! Exercise #5

1. Find and Run the **VizQL Data Service** **Queries** workflow from the **Sample Workflows** folder

We have the docs!

Information about the new **VizQL Data Service API**, and where to find them.

- [Tableau VizQL Data Service Documentation](#)
- [VizQL Data Service: Extend Your Data Beyond Visualizations](#)
- [TC24 - Infuse Data Everywhere with VizQL Data Service](#)
- [VizQL Data Service from Tableau: Use Your Data, Your Way](#)
- [Tableau VizQL Data Service \(VDS\) Query Builder](#)
- [Tableau API Collection on Postman](#)

https://github.com/Tab-SE/TC25-HOT-Postman

Tab-SE / TC25-HOT-Postman

Type to search

CodeIssuesPull requestsActionsProjectsSecurityInsights

TC25-HOT-PostmanPublic

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main1 Branch0 Tags

Go to fileAdd fileCode

jeremymayoimagesb813176 · yesterday5 Commits

images	images	yesterday
README.md	Initial commit	yesterday
TC25 Lab Instructions.pdf	lab instructions	yesterday
TC25_Postman.zip	postman collection	yesterday

README

TC25-HOT-Postman

Lab files for the TC25 HOT Session 1942 - Build, Automate & Scale: A Tableau REST API Deep Dive

About

Lab files for the TC25 HOT Session 1942 - Build, Automate & Scale: A Tableau REST API Deep Dive

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Releases

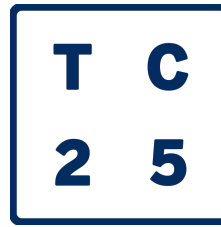
No releases published

Packages

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Sessions



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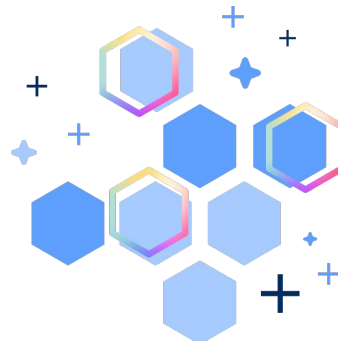
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Thank you

Coffee's on us.

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